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PREFACE

A NEED for working glass is frequently encountered in scientific research and teaching, particularly in the field of physical chemistry. In many laboratories this problem is solved by the employment of professional glass-blowers, and the research worker requires little or no skill in the manipulation of glass. Or it may be that a laboratory has no glass-blower and the scientist has to rely on his own resources, combined perhaps with the services of some outside specialist. In other laboratories the scientist may find the available services to be deficient in various respects, or overloaded, so that more satisfactory progress is made when he himself becomes an amateur glass-worker. This possibility was emphasized by W. A. Shenstone in 1889, when he wrote that the amateur with practice can make almost all the apparatus he needs for lecture or other experiments with a considerable saving in expense 'and, which very often is more important, without the delay that occurs when one depends upon the professional glass-worker.' This latter advantage is, he writes, for himself a very weighty one.

We became interested in working glass ourselves when engaged in various researches in the Chemical Laboratories at King's College, London, and over the last few years we have become increasingly convinced that the scientific glass-worker can use methods which differ from those of the professional. The scientist is primarily interested in apparatus which gives him the results he seeks for, and if this apparatus lacks elegance in appearance and is made by methods which are looked upon unfavourably by the professional glass-blower, it by no means follows that the apparatus is defective from a scientific point of view.

We have therefore compiled this work in the hope that it will be useful to scientists faced with problems of glass-working. By avoiding the more difficult manipulations involved in professional glass-working, it seems possible for a scientist to assemble quite complex apparatus, including, for example, his own mercury diffusion pumps, mass spectrometer tubes, molecular beam generators and silica fibres; the preliminary practice required, which depends upon intricate ability, may take some hours a week for a period of two or three months. We have also endeavoured to bring together data

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scattered in the literature, and to give an account of some of the more difficult methods of manipulating glass.

The researches which led to our interest in glass-working were supported by grants from the Research Fund of the University of London, from the Institute of Petroleum and from the Department of Scientific and Industrial Research (maintenance grants to A. J. C and J. D.).

December, 1956

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Chapter 1

INTRODUCTION

Glass has been known to mankind for some thousands of years. The manufacture and working of glass became a stable industry in Egypt at the beginning of the 18th dynasty. W. M. Flinders Petrie (1909) has described how, about 1370 B.C. in the time of Amenhotep IV, in the works at Tell-el-Amarna, a lump of glass was patted into a cylinder and then rolled into a rod which was drawn into cane about $\frac{1}{8}$ in. thick. This cane was wound on a mandrel to produce hollow vases. Later the art of pressing glass into open moulds was discovered. The blowpipe—an iron tube 4–5 feet long with a mouth-piece at one end—was probably discovered about the beginning of the Christian era, and, according to G. W. MOREY (1938), caused an industrial revolution. The use of tongs for manipulating glass was known to the Romans in A.D. 300. By this time, therefore, several of the basic methods which are now useful in constructing complex scientific apparatus had been developed. The glass in those early times was similar in many respects to a modern soda-lime-silica glass. Scientific method does not seem to have influenced glass production very much before the present century, although of important early investigations we may mention those of K. W. Scheele and A. L. Lavoisier on the durability of glass exposed to water and weak acids, and those of W. V. Harcourt and M. Faraday on the production and properties of glass.

The value of glass as a laboratory material is very great. A moderately skilled worker can fabricate complex glass equipment using simple tools; and perhaps of even greater value in research is the ease with which complex glass apparatus can be modified and added to with little or no dismantling. Glass is sufficiently chemically inert for most purposes, and vitreous silica may be used when extraordinary inertness is required. The transparency of glass is often valuable. Glass is a good electrical insulator, and metal electrodes in a glass envelope can be raised to incandescence by eddy-currents induced by a coil, outside the envelope, carrying high-frequency current. In vacuum researches glass is valuable because of the ease with which leaks are found with a Tesla coil, and on account of its

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ready outgassing on baking. For nearly all practical purposes glass is vacuum-tight. Glass-to-metal seals of various kinds may be made, and different types of glass can be sealed together with appropriate intermediate glasses.

The manipulation of glass is a craft and has been handed down over the centuries largely by personal example and tradition. Venice was the most important centre of glass-working for the four centuries following its rapid development in the eleventh century to a dominating position; in 1279 a system of apprenticeship needing eight years was introduced there, and the closed and somewhat secretive nature of the craft was reinforced by the employment of assassins to remove glass-blowers who seemed likely to give away valuable secrets to other centres of the industry. Perhaps the aura of mystery which still to some extent surrounds scientific glass-working can be traced back to these times.

The basic techniques developed for scientific glass-working involve the manipulation of glass in the form of tubing or rod, using a flame as a source of heat. The article to be fashioned is held by the operator in both hands, and the part to be worked becomes a semi-molten mass in the flame and is shaped principally by blowing. Generally it is necessary to rotate the article in the flame, and this often requires a very high degree of muscular co-ordination, which can be developed by professional workers to a remarkable extent. This rotation operation is not easy; M. FARADAY (1842) stated the outstanding difficulty in the following words: 'But when the heat has brought the glass into a soft state, it is by no means easy so exactly to turn the tube at both ends alike, and so lightly yet equally to hold them, that the soft part shall retain its cylindrical shape; being neither twisted, nor bent, nor elongated, nor thrust up.' R. E. THRELFALL (1946) considers that one third of the art of glass-blowing consists essentially in being able to move both hands about, rotating a tube with each finger and thumb, and keeping constant both the distance between the hands and the speed of rotation. Considerable practice is necessary to gain great mastery of this technique, which we can call the bench-flame method of work, since the flame is in a fixed position on the bench. The problem facing the scientist who has to engage in his own glass-working is that of simplifying or modifying those operations which depend on extremely good muscular co-ordination. We consider this problem to be soluble when elegance can be sacrificed to utility.

The requirements of a scientist differ from those of a professional glass-worker. The starting point of a new research is new ideas, and when these require subsequent experimental investigation the apparatus used need be only good enough to give results which are

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satisfactory from a scientific point of view. A glass apparatus must, in fact, hold together and work, but it may have an unsightly appearance. However, we must also note that experimental skill in itself is of great value to the scientist in pursuing his ideas, for as FARADAY (1842) wrote: 'The person who could devise only, without knowing how to perform, would not be able to extend his knowledge far, or make it useful; and where the doubts or questions that arise in the mind are best answered by the results of an experiment, whatever enables the philosopher to perform the experiment in the simplest, quickest, and most correct manner cannot but be esteemed by him as of the utmost value.' In addition to these considerations, experimental skill is of particular value in developing new techniques; and these may lead to many unexpected developments. H. E. ARMSTRONG (1924) tells how the introduction in 1865 of the Sprengel pump—a fairly simple piece of glass-working—revolutionized the English water supply. Again we may note the important practical consequences following on the early work of H. L. CALLENDAR (1892) on vitreous silica tubes, and that of C. V. BOYS and R. THRELFALL on vitreous silica fibres.

In view of the difference in objective between the scientist and professional glass-worker, we consider that glass-working for the scientist should develop as its own craft, and indeed this has happened to a certain extent. An early systematic account is that of J. J. BARZELIUS (1833), which describes some of the basic operations. FARADAY'S *Chemical Manipulation* is a masterly treatment, referred to frequently in R. THRELFALL'S notable work *On Laboratory Arts* (1898), which devotes 107 pages to glass-blowing and manipulation of glass. Threlfall seems to have been one of the first to describe what is now often called 'in-place' glass-blowing, in which the glass pieces are kept stationary by clamps, and the flame is moved. Joints are made with thin glass rod, now often called welding rod, by a method very similar to that used in 1370 B.C. in the Tell-el-Amarna factory. The difficulty of holding and rotating the glass is avoided, and thus, as Threlfall says, the method is most useful to the experimenter who wants to get on to other things before sufficient skill is acquired for the rotation method. The tradition of the craft of glass-working for scientists is continued, we think, by the works of W. A. SHENSTONE (1889), T. BOLAS (1898), B. D. BOLAS (1921), F. C. FRARY, C. S. TAYLOR and J. D. EDWARDS (1928), R. H. WRIGHT (1943), and J. D. HELDMAN (1946). Amongst eminent scientists who have carried on the glass-working craft with their own hands we may mention R. W. BUNSEN, whose skill at the oil-fed blowpipe and whose exemplary patience when one of his pupils rapidly and several times

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in succession broke the apparatus glass-blown by the master himself are described by H. E. ROSCOE (1901); J. Dewar, whose vacuum-jacketed glass vessels marked a new era in low temperature research; M. Bodenstein and W. Ramsay, both of whom are described by E. K. RIDGAL (1951) as uncannily skilled in the art of glass-blowing; and F. W. Aston, whose glass bulb discharge tubes and cooled charcoal-in-glass pumping systems first gave those exact mass relations of isotopic chemistry which contributed decisively to the opening of the atomic age. In modern times the old tradition that the masters of science should also be themselves masters of the practical craft of science has failed to persist, and the more eminent scientists are now usually not to be found at the working bench in the research laboratory. There is, we think, a consequent weakening of the craft basis of practical scientific work, and some retreat from the view expressed by I. Langmuir that work in the laboratory can be fun.

The increasing use in scientific research of borosilicate glasses over the last thirty years has not, on the whole, been accompanied by much departure from the traditional methods of glass-working. Generally it is much easier to rotate the flame about a stationary clamped article than to rotate the article in a stationary flame. Quite complex apparatus can be constructed by working with a hand torch—giving a movable flame held in the hand—on clamped apparatus. This method of work is specially suitable for the borosilicate glasses of low thermal expansion, which can be worked into quite knobby apparatus without there being much danger of cracking on cooling—in contrast to soda-lime-silica glass. The joining of two clamped tubes with a movable flame has been described by SHENSTONE (1889), THRELFALL (1898) and TRAYNERS (1901). A. L. REIMANN (1952) has described some further uses of this general method, but the great number of complex operations which can be carried out with facility using a hand torch in place of a bench torch does not seem on the whole to have been realized; indeed, HELDMAN writes: 'End-seals with both tubes of approximately the same diameter and T-seals are, with practically no exceptions, the whole repertory of in-place glass blowing.' We are by no means in agreement with this statement. Many operations can be carried out with a hand torch on completely fixed glass, or on a fixed piece of glass to which some other part of glass can be joined by holding it in one hand whilst the other hand manipulates the hand torch. The results are usually not as elegant as those obtained by a skilled worker using rotation in a bench torch, but less skill and practice are required for the hand torch method. It has the further great advantage that the complexity of the apparatus being constructed can be steadily increased without

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greatly increasing the glass-working difficulty. Furthermore, the scientist will probably only work glass from time to time, depending upon the requirements of the research or other work, and the hand torch method is far more suitable than the bench torch method for the operator who does not want to spend time on preliminary practice; it is also far more suitable for the scientist working in awkward positions on complex apparatus. We find that very complex apparatus may be assembled with a hand torch.

A clear account of professional methods of glass-working has been given by W. E. BARR and V. J. ANHORN (1949). Valuable information on some glass-working problems is also found in the works of J. BRONG (1940), M. C. NOKES (1948), A. J. ANSLEY (1950), A. ELLIOTT and J. HOME DICKSON (1951), and H. J. J. BRADDICK (1954). Accounts of the simpler operations have been given by W. E. PARK-WINDER (1947) and E. H. MORGAN (1953).

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Chapter 2

PHYSICAL AND CHEMICAL PROPERTIES OF GLASS

The Structure of Glass

A glass is a product of fusion which has cooled to a rigid condition without crystallizing. This definition includes a large number of organic glasses, and does not restrict the term 'glass' to inorganic substances, which is a frequent practice in the U.S.A. This restriction seems somewhat arbitrary, particularly when we consider how G. Reimann established the general principles of the glass-like state by research on organic glasses, and how the devitrification of technical bases is paralleled by that of organic glasses. Tammann concluded that a glass could be regarded as a supercooled liquid in which the rotational movements of the molecules had been frozen (see W. E. Hamer, 1952). In fact, as R. Boyle described it about 1660, 'the particles of the glass agitated by the heat, were surpriz'd by the cold before they could make an end of those motions which were requisite for their disposing themselves into the most durable texture.' In modern terminology, a glass is thermodynamically unstable with respect to the corresponding crystal.

Crystalline silica (quartz, tridymite or cristobalite) in its various modifications is built up of SiO_4 tetrahedra linked together in a regular manner so that every oxygen is between two silicons. The tetrahedra therefore share corners. The arrangement in space of the tetrahedra is different in the various crystalline forms, but is always fairly regular. A silica glass, in contrast, again contains SiO_4 tetrahedra with every corner shared; but by slight distortions of the valency angles, as compared with the crystal, a continuous and irregular three-dimensional network is built up. The orientation about the Si-O-Si bond of one SiO_4 tetrahedron with respect to another can be practically random. Thus a two-dimensional picture of a silica glass would show a series of irregular rings, with an average number of about six tetrahedra in each ring, but with the number of tetrahedra in individual rings varying from three to ten or more. The silica glass satisfies the condition for glass formation proposed by W. H.

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ZACHARIASEN (1932), namely that the substance can form extended three-dimensional networks lacking periodicity, with an energy content comparable with that of the corresponding crystal network. A glass does not, therefore, produce a regular diffraction pattern with x-rays; but a monochromatic x-ray beam incident on a glass is scattered, and a radial distribution curve may be constructed. The space average of the distribution of atoms round a given one can be deduced (see J. T. RANDALL, 1938). Much work of this kind has been carried out by B. E. WARREN and his colleagues.

A soda-silica glass results from the fusion of Na_2O with SiO_2 . The number of oxygens is more than twice the number of silicones, and some of the oxygens are bonded to only one silicon. A silicon bonded to one of these oxygens is at the centre of a tetrahedron which shares only three corners with other tetrahedra. With each singly bonded oxygen there is associated one negative charge. The sodium ions are found in the holes in the three-dimensional silicon-oxygen network. On the average, each sodium is surrounded by about six oxygens, and each silicon by four oxygens. In a soda-boric oxide glass of low soda content the extra oxygen is bonded between two borons, and there are no singly bonded oxygens. This can happen because in a boric oxide glass the co-ordination of boron by oxygen is triangular, and in the mixed glass some of the boron atoms become tetrahedrally co-ordinated by oxygen. When there is more than about 13-16 per cent of Na_2O in the glass, the boron atoms cease to change their co-ordination, and some singly bonded oxygens exist (B. E. WARREN, 1942).

Soda-silica glasses are not formed when the soda content exceeds that given by the formula Na_2SiO_3 . For this formula, if every silicon atom is surrounded tetrahedrally by four oxygen atoms, then on the average two oxygens round every silicon are singly bonded, and a continuous network is just possible. With still more oxygen it is not possible. In a soda-silica glass with much less soda, there are many SiO_4 tetrahedra sharing every corner, and a number sharing only three corners. The way in which these different tetrahedra are distributed is not yet quite clear. There may be small regions where all the tetrahedra share four corners, and such regions are composed of pure silica; they may alternate with regions of, for example, $\text{Na}_2\text{O} \cdot 2\text{SiO}_2$. The composition may vary through the glass when sufficiently small regions are considered.

The general picture of a glass as a negatively charged irregular framework containing holes with positive ions in them enables a distinction to be made between the network-forming ions, which comprise the framework, and the network-modifying ions which go

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in the holes. Silicon, boron and phosphorus are important network-forming ions. Sodium and potassium are important network-modifying ions. Other ions can act in both capacities. This is probably true of aluminium, beryllium, zinc, iron and titanium. In a soda-silica glass it seems that lead atoms can take part in the network and link SiO_4 tetrahedra together. Cobalt ions in network-modifying positions tend on heating to move into the network, and this can cause a colour change from pink to blue.

Most commercial glasses are based on silicates or borosilicates. A typical hard borosilicate glass for chemical work may contain 80 per cent SiO_2 , 12 per cent B_2O_3 and 4 per cent Na_2O . A soft soda-lime-silica glass (usually referred to as soda glass) may contain 70 per cent SiO_2 , 17 per cent Na_2O and 5.4 per cent CaO . Lead glasses, used for lamp and valve stems, may contain 30 per cent PbO , 57 per cent SiO_2 , 5 per cent Na_2O and 7 per cent K_2O . These glasses have high electrical resistance. Glasses with exceptionally high softening temperatures contain 20-25 per cent of Al_2O_3 . Borate glasses, substantially free from silica (8 per cent SiO_2) are used for sodium discharge lamps.

General accounts of the structure of glass have been given by J. E. PRANWORTH (1950), B. E. WARREN (1940) and C. J. PHILLIPS (1948).

General Physical Properties of Glass

The physical properties of a given specimen of glass may depend upon the previous history of the specimen. This is particularly the case for the mechanical strength under tension, when the surface preparation of the specimen is of decisive importance. The thermal expansion and viscosity of glass also depend to some extent on the history of the specimen. The importance of this factor has been emphasized by A. E. DALE and J. E. STANWORTH (1945). R. W. DOUGLAS (1945) has given a valuable review of the physical properties of glass.

Mechanical Strength

The important property for the practical worker is the strength of glass under tension. The surface of glass very probably contains numerous extremely small cracks extending into the glass, and when tensile stress is applied there is a concentration of stress at the ends of these cracks, which causes them to grow further into the glass, until at some crack breakage occurs and is propagated through the specimen. Glass usually breaks in a direction at right angles to the direction of maximum tensile stress. A newly-drawn glass fibre is free from these surface cracks and is much stronger than an old

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fibre; but touching a new fibre, even with the fingers, greatly weakens it. An old fibre is actually strengthened by removing the surface layer with hydrofluoric acid, even though the cross-section is reduced. The strength of a glass under tension varies from one specimen to another. A further complication is the variation of the tensile strength with the time for which the stress is applied. A tensile stress which does not cause fracture after a short time of application may do so after a long time. There is in fact a delayed fracture of glass.

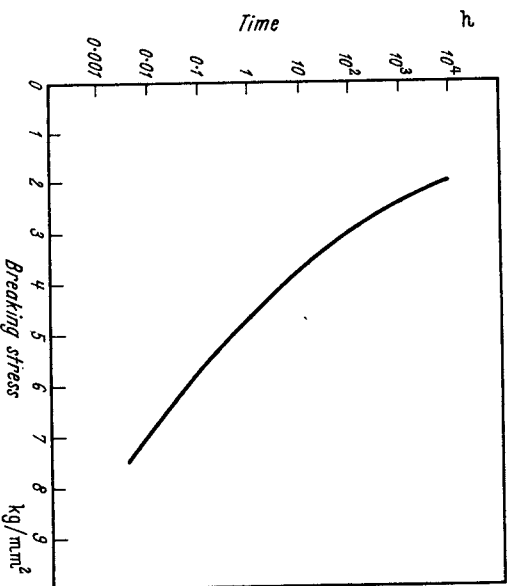


Figure 1. Nature of the relation between time of loading and breaking stress for glass. This property is of a statistical nature, and the particular curve shown can only be taken as representative

The nature of the relation between time of loading and breaking stress for a borosilicate or soda-lime-silica glass is shown in FIGURE 1. The curve given must be taken as representative only. A typical figure for the safe tensile strength for prolonged loading times is 0.7 kg/mm² (1000 lb/in.²). Similar results for the relation between bending stress and time to fracture are found when a tensile stress is produced by bending a glass rod into an arc of a circle. From FIGURE 1 we note that an increase of stress by a factor of 4 reduces the time required for fracture by over 10⁶ times. If a certain load is supported for one hour by a certain piece of glass, one quarter of the load should be supported for a million hours. This can be made use

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of in testing the extent to which stress can be applied to a glass apparatus. Four times the stress the glass must support can be applied for a short time.

The delayed fracture of glass, shown in FIGURE 1, must be borne in mind in assembling apparatus. If, for example, an apparatus is clamped so that bending stress is introduced, the apparatus may break after a long interval. Chemical reactions at the surface of the glass may be partially responsible for delayed fracture: C. GURNEX and S. PEARSON (1952) found a soda-lime-silica glass to be stronger in vacuum, and to be weakened by carbon dioxide and water in the surrounding atmosphere.

Thermal Expansion

The coefficient of linear thermal expansion is almost constant, for most types of glass, for temperatures up to 400–600°C. The actual value depends on the chemical constitution of the glass. It then increases rapidly above a certain temperature, often called the

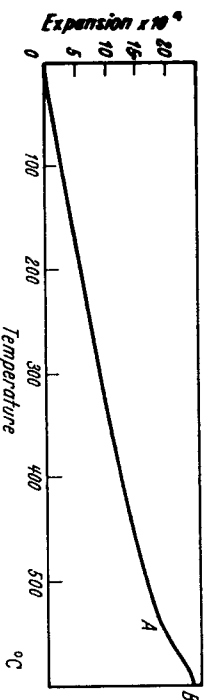


FIGURE 2. A typical expansion curve for a hard borosilicate glass (Phoenix)

'transformation point'. This is not, however, a characteristic temperature, since it depends on the thermal history of the specimen and the rate of heating. At a higher temperature the glass softens and ceases to expand. This is sometimes called the 'softening temperature' or the 'Mg point'. Confusion may result from another definition of softening temperature, depending upon the rate of extension of a fibre by viscous flow. This latter softening temperature, which corresponds to a viscosity of 10^{7.6} poises, is very much higher than the Mg point. A typical linear expansion curve for a borosilicate glass (Phoenix) is shown in FIGURE 2. A is the transformation point and B the Mg point. The temperature corresponding to A is often called the 'lower annealing temperature', and corresponds to a viscosity of about 10¹⁴ poises; that corresponding to B is often called the 'upper annealing temperature', and corresponds to a viscosity of

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about 10^{12} poises. Another definition in common use, especially in the U.S.A., is to call the 'annealing temperature' that at which the viscosity is $10^{13.4}$ poises. This is then between *A* and *B* on the thermal expansion curve. It is useful to bear in mind the confused state of terminology and definitions when using tabulated data on the thermal properties of glass. The viscosities mentioned above are, perhaps, not established with certainty. At the lower annealing temperature, annealing is actually extremely slow. This temperature is not used for the practical annealing of laboratory apparatus.

Thermal Endurance

This measures the ability of the glass to stand sudden changes of temperature without fracture. When a specimen of glass is suddenly heated uniformly over all its surface, the heat penetrates slowly into the interior. The outside layers are heated first, and being unable to expand fully they become subject to a compressive stress, whilst the inner layers become subject to a tensile stress. When the specimen at a uniform temperature is suddenly cooled over all its surface, the surface layers are subject to tensile stress. Since the mechanism of fracture usually involves surface cracks, glass is more likely to break on sudden cooling than on sudden heating. The magnitude of the stress produced on sudden cooling depends on the modulus of elasticity and the coefficient of linear thermal expansion, and, in a way not important in practice, on Poisson's ratio. Thermal endurance is measured by somewhat empirical methods, and is again a statistical quantity. A heat-resisting glass is one having a high thermal endurance; a hard glass has a high softening temperature. A 1-mm thick beaker of a hard borosilicate glass, such as *Pyrex*, *Phoenix* or *Firminasil*, will require a thermal shock, by sudden cooling, of about 325°C to give appreciable probability of fracture. For a soda-lime-silica beaker the corresponding figure is about 120°C . Beakers of *Monax* glass stand a much greater thermal shock than the soda-lime-silica beaker; the beakers of standard thickness can usually survive a thermal shock of 240 – 250°C . Thick glass fractures with less thermal shock than thin glass.

The glass-worker subjects tubing to thermal shock by suddenly placing it in a hot flame. The inner surface of the glass tube is then not heated directly, and is very quickly subjected to tensile stress. The hard borosilicate glasses as tubes can usually be placed immediately in an oxy-coal gas flame without fracture, but complex apparatus, especially when internal seals are present, requires more gentle heating. Soda-lime-silica glass tubes need gentle warming at first, particularly when the end of a tube which has not been fire-polished is put in the

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flame. The end of a tube is fire-polished by fusing it in a flame, and this process closes up surface cracks. Vitreous silica has very great thermal endurance: small red-hot articles can be quenched in water without cracking.

Thermal Conductivity

For the hard borosilicate glasses and the soda-lime-silica glasses this is around $0.0025 \text{ cal } ^{\circ}\text{C}^{-1} \text{ cm}^{-1} \text{ sec}^{-1}$. For vitreous silica (*Vitresil*) in the transparent form it is 0.0025 up to 500°C , and 0.0035 from 500 to 1000°C ; for the translucent form it is 0.0033 .

Viscosity and Softening Temperatures

These properties have already been mentioned in connection with thermal expansion. The viscosity decreases rapidly with increasing temperature. A linear relation is found between the logarithm of the viscosity and the reciprocal of the absolute temperature. This is convenient for extrapolation. When the viscosity has the value $10^{17.6}$ poises the glass is mobile enough to be drawn into threads, and the temperature is sometimes called the softening temperature (see p. 11). At temperatures between the lower and upper annealing temperatures (*A* and *B* in FIGURE 2) the viscosity can change with time—when the glass is suddenly cooled the viscosity slowly increases to an equilibrium value and when the glass is heated the viscosity slowly decreases to an equilibrium value—in fact time is required for the equilibrium viscosity values to be attained. Glass is often worked when its viscosity is about 10^4 poises; for a hard borosilicate glass this corresponds to a temperature of about 1200°C .

Elastic Properties

Pyrex Chemical Resistance Glass has a Young's modulus of 6.1×10^{11} dynes/cm², a modulus of rigidity of 2.5×10^{11} dynes/cm² and a Poisson's ratio of 0.22. Similar values are found for other glasses.

The extension of an amorphous material under a tensile force can be resolved into three parts; first, an immediate elastic extension, which is immediately recoverable on removing the tensile force; and secondly, a delayed elastic extension which is recoverable slowly; and thirdly, a plastic extension, viscous flow, or creep, which cannot be recovered. With glass at ordinary temperatures, this plastic extension is practically absent. A very slow delayed elastic extension occurs. This effect can be troublesome in work with torsion fibres. The delayed elastic effect in vitreous silica fibres is 100 times less than in other glass fibres, and viscous flow of silica is negligible below 800°C (N. J. TIGHE, 1956). For exact work vitreous silica torsion fibres are therefore used.

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Thermal Capacity

For Pyrex glass the thermal capacity (specific heat) between 0 and 250°C is given by $C \text{ (cal gm}^{-1}\text{°C)} = 0.174 + 0.00036t$ where t is the temperature in °C.

Electrical Resistance

The resistance of vitreous silica (*Vitreosil*) in the translucent form at room temperature exceeds 2×10^{14} ohm cm. Glasses containing metal ions in network-modifying positions are ionic conductors. In a soda-lime-silica glass, for example, the current is carried by sodium ions and the resistance at 150°C may be around 10^8 ohm cm. *Lemington W.1.*, a hard borosilicate glass, has a resistance of about 10^{10} ohm cm at 200°C. A typical lead glass, *Wembley L.1.*, has a very much greater resistance both at room temperature and normal lamp-operating temperatures than a soda-lime-silica glass, and is therefore valuable for lamp and valve pinches. The resistance of *L.1.* at 150°C is 10^{12} ohm cm. Generally the volume resistance due to ionic conduction decreases rapidly with temperature. The logarithm of the conductivity is a linear function of the reciprocal of the absolute temperature. The surface of most glasses is very hydrophilic, and there is a surface conductivity which depends upon the relative humidity. For *Phoenix* glass, for example, the volume resistance of a centimetre cube at room temperature is about 3×10^{14} ohm, but the surface resistance at 60 per cent relative humidity is 7×10^{11} ohm, and at 81 per cent relative humidity it is 5.4×10^9 ohm. In very humid atmospheres it is possible to have an electrical shock by touching the surface of a soda glass apparatus containing electrodes at high potential. The water layer on the glass becomes slightly alkaline after a time by reaction with sodium from the glass; the apparatus should be wiped from time to time with a cotton cloth. It is best in these cases to use a borosilicate glass. The surface conductivity of glass was discovered by M. FARADAY (1830).

Density

For the soda-lime-silica glasses this is about 2.5 gm/cm³; for the borosilicate glasses it is very nearly 2.25 gm/cm³ and hardly changes with slight variations in composition. *Wembley L.1.* lead glass has a density of 3.08. A very dense lead glass has a density of 5.2.

Hardness

Generally glasses with a high silica content are more resistant to abrasion than low silica content glasses. The hardness therefore increases with increase of softening temperature. Lead glasses can be scratched quite easily.

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Transmission of Light

A 1-mm thick sheet of *Phoenix* glass will transmit 90 per cent or more of the light incident on it, for wavelengths of 350 millimicrons to almost 2 microns. In the infra-red region a strong absorption occurs at 3 microns and little transmission beyond 4 microns. In the ultra-violet region increasing absorption occurs as the wavelength falls below 350 millimicrons and very little transmission occurs below 270 millimicrons. The transparent variety of vitreous silica (fused quartz) has very superior optical properties, and is widely used in photochemical and optical researches. In the ultra-violet region it transmits at high efficiency down to 1850 Ångström units (185 millimicrons). 'Quality O.H. *Vitreosil*' of The Thermal Syndicate Ltd is a special optical quality in which the absorption band at 2400 Ångströms has been eliminated. A special quality of fused quartz is also available which transmits infra-red up to 3.5 microns approximately ('I.R. quality *Vitreosil*' of The Thermal Syndicate Ltd). In this *Vitreosil* the absorption band at 2.7 microns has been much reduced.

The Stress-Optical Coefficient

It is not usual to take quantitative measurements of the strain in glass apparatus made for research; when a strain-viewer is used (p. 43) qualitative observations are normally made. Quantitative measurements can be made when the stress-optical coefficient is known. The theory involved requires a knowledge of the optical behaviour of doubly refracting materials and depends on the fact that a ray of plane-polarized light entering strained glass is broken into two rays—the 'ordinary ray' and the 'extraordinary ray'—vibrating at right angles to each other. For glass subject to simple axial tension or compression, the extraordinary ray vibrates in the plane which includes the axis of the stress. The birefringence of strained glass is proportional to the strain, and thus to the stress. The stress-optical coefficient is the maximum double refraction or birefringence observed in polarized sodium light for 1 cm path length when there is a uniform stress of 1 kg/cm². It is expressed either in wavelengths of sodium light or in millimicrons. This coefficient varies from one glass to another; it is around 3.5 millimicrons, or 0.006 wavelengths of sodium light. A. JOHANSEN (1918) has given an account of methods for determining double refraction, and very valuable data for practical work are given by J. H. PARTRIDGE (1949).

General Chemical Properties of Glass

Resistance to Chemical Actions

Vitreous silica is the most chemically inert glass for most purposes.

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It is not affected by halogens or acids, except for phosphoric and hydrofluoric acids. Phosphoric acid attacks fused silica at temperatures of 300–400°C, and hydrofluoric acid attacks it at room temperature, forming silicon tetrafluoride and water. At high temperatures silica reacts with caustic alkalis, certain metallic oxides, and some basic salts, and cannot be used for incinerating these materials. Over 1600°C, fused silica is reduced to silicon by carbon. It can also be reduced at high temperature by hydrogen. It is unaffected by water under normal conditions but is attacked by strong solutions of alkalis.

The hard borosilicate glasses are highly resistant to attack by water; but just as the sodium ions in the glass are slightly mobile under the influence of an electric field (p. 14), so also they can be mobile by thermal agitation and escape from the glass into water in contact with it and be replaced by hydrogen ions. This effect is slight: for example, a *Firmasil* beaker in an autoclave containing water at 150°C loses about 0.00015 gm of sodium per dm² in four hours. A soda-lime-silica glass loses sodium to water at a much greater rate. The resistance of borosilicate glass to most acids is very good, but strong aqueous alkalis produce visible attack. The network of triangles and tetrahedra is attacked, so the glass tends to dissolve as a whole. Soda-lime-silica glass usually has less chemical resistance than a borosilicate glass. Alkaline attack, however, becomes much greater on glasses with high silica content. Alkalis can also leach out boric oxide from a borosilicate glass. Hydrofluoric acid dissolves glass, and glacial phosphoric acid attacks most kinds of glass.

The Weathering of Glass

A reaction between sodium from the glass and atmospheric water and carbon dioxide can lead to the formation of sodium carbonate, which crystallizes in fine needles. A potash glass forms potassium carbonate, which is too deliquescent to crystallize out. A lead glass can react with hydrogen sulphide, and to a smaller extent with carbon dioxide, sulphur dioxide, and acid vapours.

Phenomena Arising from the Heating of Glass

A rapid evolution of adsorbed water first occurs on heating glass; this is followed by a persistent evolution, due to gas (mostly water) diffusing from the interior. Above 300°C the two processes are fairly clearly separated. The adsorbed water is rapidly and completely removed, and the quantity of gas evolved by the persistent evolution

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is proportional to the square root of time. The process has an activation energy. For a soda-lime-silica glass over 98 per cent of the evolved gas is water. B. J. TODD (1955) has studied these effects in detail. The adsorbed water on glass can be troublesome in gaseous manipulation, as R. W. BUNSEN first appreciated.

At high temperatures glass loses its more volatile components. The loss of silica, lime, magnesia and alumina is negligible, but boric oxide, lead oxide, sodium oxide and potassium oxide can also be lost. When the glass is heated in a flame, reaction may occur with some of the flame gases; sulphur dioxide can react with soda glass and lead glass to form sodium sulphate and lead sulphate respectively, and of these only the former can be washed off. An account of these effects is given by W. E. S. TURNER (1945). The loss of weight of vitreous silica on ignition is negligible; crucibles can be heated to 1050°C, and precipitates can be ignited at 1000°C in crucibles with a porous base of vitreous silica.

Diffusion through Glass

The mobility of the sodium ions in a soda-lime-silica glass at elevated temperatures is fairly high; if an evacuated bulb of such a glass is dipped into molten sodium nitrate and electrolysis is brought about by bombarding the inside of the bulb with electrons, the circuit being completed with an electrode in the sodium nitrate, then metallic sodium appears in the bulb. By immersing the bulb in other molten salts the sodium ions can be replaced by ions of silver, copper, thallium and vanadium. These ions also diffuse into glass from their molten salts in the absence of an electric field. When potassium is distilled in a borosilicate glass vessel it becomes slightly contaminated with sodium which diffuses from the glass and is replaced by potassium (D. K. C. MACDONALD and J. E. STANWORTH, 1950). Vitreous silica allows helium, hydrogen, neon, nitrogen, oxygen and argon to diffuse through it, with the permeability decreasing in the order given. The permeability of silica becomes greater if the glass devitrifies. The permeability to helium of soda-lime-silica glass is 10⁵ (or more) times less than that of vitreous silica. For practical vacuum purposes soda and borosilicate glasses can be regarded as impermeable to gases at ordinary temperatures, except in work at extremely low pressures when the diffusion of atmospheric helium through the glass may become significant.

The permeability of glass at high temperatures seems to have been discovered by R. BOYLE. In his collected works published in 1744 there is a paper in Volume III 'A discovery of the perviousness of glass to ponderable parts of flame' in which he writes '... it is plain

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that igneous particles were trajected through the glass, which agrees with the Epicureans . . .'

Some Surface Properties

The chemical properties of glass are largely determined by its surface, because the metal ions present in most types of glass are in fact reactive enough, but only those at the surface can react. The very slow rates of diffusion in glass at ordinary temperatures limit its reactivity, so that, as M. FARADAY (1830) wrote 'Glass may be considered rather as a solution of different substances one in another, than as a strong chemical compound; and it owes its power of resisting agents generally to its perfectly compact state, and the existence of an insoluble and unchangeable film of silica or highly silicated matter upon its surface.' The surface composition, for volatilization occurs during the forming process, and weathering occurs subsequently; both processes produce a surface resembling vitreous silica. It is possible that the SiO_4 tetrahedra on the surface terminate in OH groups to which adsorbed water is normally bound. The glass surface can be made hydrophobic instead of hydrophilic by allowing adsorbed water and surface hydroxyls to react with various monoalkyl/dichlorosilanes (RHSiCl_2), when hydrogen chloride is formed by elimination.

The surface properties of glass are of great importance in many reaction kinetic studies, particularly those involving the termination of reaction chains on the walls of the vessel. When a glass reaction vessel is used in such cases, it is usually found that it must be matured, by carrying out a number of reactions in it, before reproducible results can be obtained. In many cases, reproducible results are only obtained when the vessel is kept continuously at the reaction temperature, and exposed only to the reaction mixtures; if the apparatus is cooled, and air let in to make an alteration or repair, different results may be obtained subsequently. This is particularly the case in oxidation reactions.

Devitrification

This is the process of the crystallization of one or more of the constituents of glass. Generally a glass is thermodynamically unstable with respect to these crystals, but at ordinary temperatures the crystallization rate is quite negligible. Crystallization may occur when the glass is worked at high temperature. The crystals which appear in a supercooled melt are not necessarily those of the stable solid phase at the temperature concerned: for example, cristobalite can appear at temperatures for which tridymite is the stable crystalline

DEVITRIFICATION

form of silica. The crystals which are most likely to separate from soda-lime-silica glass are those of calcium silicate, tridymite and cristobalite. Calcium silicate occurs usually as the monoclinic form wollastonite, but sometimes in a hexagonal form. The monoclinic form tends to appear as very long, thin crystals bunched together to give a brush-like appearance. The hexagonal form is not acicular. Tridymite is hexagonal, and forms hexagonal stars, like snow, by twinning at 60° . The cristobalite forms twins at 90° . Devitrification on the surface of glass seems to depend upon loss of alkaloxides, and may be assisted by dust particles.

Transparent vitreous silica (transparent *Vitreosil*) is liable to devitrify if potassium or sodium compounds are present. The surface of the material must be thoroughly cleaned, and the part to be worked should then not be touched with the fingers.

When devitrification is observed in the soda and borosilicate types of glass, the crystals may sometimes be removed by fusion of the glass in the flame; alternatively the semi-molten devitrified portion may be removed with tongs, and replaced by fresh glass added as thin rod. If there is a large extent of devitrification the portion of glass should be completely cut out and replaced. Old soda glass apparatus is very liable to devitrify when repaired: F. C. FRARY, C. S. TAYLOR and J. D. EDWARDS (1928) advise, in such cases, a preliminary washing of the glass with dilute hydrofluoric acid to remove the surface layer.

Thermal Strain in Glass

When a block of glass is suddenly heated on all its faces, the outer layers are under compression and the inner layers are under tension, as explained on p. 12. This strain is temporary, however, since it vanishes as soon as the temperature gradient vanishes. Temporary strain is similarly produced when the surface of a block of rigid hot glass is cooled. Above the upper annealing temperature (p. 11) a stress can only exist in glass for a short time, because the glass flows to relieve the stress. Stress is relieved only very slowly at the lower annealing temperature. Thus over a certain temperature range glass changes from a viscous to a rigid body. Consider a block of glass to be rapidly cooled through this temperature range, so that there is always a temperature gradient. In the viscous region the glass is strain-free, and thus when it first becomes rigid it is also strain-free. Thus there is a rigid block of glass containing a temperature gradient but free from strain. When this temperature gradient is removed, the inner layers of the glass are in tension and the outer layers in compression. There is then a permanent strain in the glass. Clearly, when

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a complex glass object is cooled rapidly there will finally be a complex distribution of permanent strain, and this can be great enough to cause fracture of the glass.

The drops of glass produced by Prince Rupert of Bavaria by dropping molten glass into oil become rigid when there is a great temperature gradient and the surface is consequently in strong compression. This makes the drops very strong, but they disintegrate violently as soon as the tail of the drop is cut off, when the internal stresses are no longer balanced.

Annealing of Glass

The object of this process is to prevent permanent strain arising from the cooling of glass. The glass must be cooled slowly through the critical temperature range in which it becomes rigid and ceases to relieve internal stresses by viscous flow. The rate at which these stresses are relieved in the annealing range of temperatures (*A* to *B* in Figure 2) depends on temperature; when this is such that the viscosity is $10^{3.4}$ poises, the glass will become practically stress-free in 15 minutes (A. E. DALE and J. E. STANWORTH, 1945). Below the lower annealing temperature the glass can be cooled quickly without introduction of permanent strain, but the temporary strain could become great enough to fracture the article.

Annealing is carried out most satisfactorily in an oven (p. 45). Complex articles of Pyrex glass can be annealed at 560°C for 30 minutes, followed by slow cooling with the oven door shut. Articles of *Firmasil* glass should be annealed at 575°C, but even at 475°C strain is very slowly removed. For *Phoenix*, the upper annealing temperature is 600°C and the lower annealing temperature is 520°C. Annealing at 560–580°C is therefore satisfactory for this glass. The annealing temperatures of these borosilicate glasses are not at all critical. The article must not, of course, be made too hot, or it will deform. Annealing is of great importance for articles made of a soda-lime-silica glass. *Wembley X.8*, soda glass should be annealed in the range 520–400°C, and the General Electric Company, which makes this glass, recommends annealing at a high temperature of 520°C for 5–10 minutes, followed by cooling to an intermediate temperature of 460°C at a rate dependent upon the glass tubing thickness. These rates are:—

3°C per minute for $\frac{1}{8}$ mm wall thickness
2°C per minute for 1 mm wall thickness
1°C per minute for 3 mm wall thickness.

The glass should be cooled from the intermediate temperature of 460°C to a low temperature of 400°C at double the above rates. The

ANNEALING OF GLASS

article can then be cooled to room temperature at any rate possible without cracking it by temporary thermal strain. The corresponding high, intermediate and low temperatures for *Wembley L.1*, lead glass are 430°C, 390°C, and 340°C. The same annealing schedule can be used. For *Wembley M.6*, 'white neutral' glass the temperature range is 580–450°C. Again the same schedule should be used.

Complex apparatus assembled on the bench must be annealed by flame, and this method must also be used when no oven is available (see p. 170). In our experience, very complex apparatus of Pyrex glass can be flame-annealed satisfactorily. Usually with Pyrex the apparatus either cracks in a day or two after making, or else not at all. Complex apparatus of soda-lime-silica glass can be annealed by flame, but we do not find this satisfactory. For research apparatus it is best to avoid this kind of glass. With a complex vacuum apparatus of borosilicate glass a fracture can often lead to unfortunate consequences, especially when there are many mercury cut-offs present; and in such cases it is well, before evacuating, to wait for a few days after a repair or alteration has been made in a position where flame annealing is difficult.

Some Types of Glass for General Use

Many different kinds of glass are made. In this Section and the following Sections we mention only a few of these which are useful in the laboratory.

Wembley X.8.

This is a soda-lime-silica glass, containing magnesia and boric oxide less than 1 per cent), made by the General Electric Co. It is often described as GEC X.8. or simply as X.8. The linear coefficient of thermal expansion between 20 and 350°C is $9.65 \pm 0.10 \times 10^{-6}$. This glass is available as tubing and rod in a wide range of sizes.

TH No. 94

Made by the British Thomson-Houston Co., this is a soda-lime-silica glass. The linear expansion coefficient is 9.5×10^{-6} (50–400°C). The glass is available as machine-drawn tubing for bench working etc.

Monax

This is a borosilicate glass, free from arsenic, antimony and lead, made by John Moncrieff Ltd. The coefficient of linear expansion is $6.4 \times 10^{-6}/^{\circ}\text{C}$. The glass is available as tubing and rod of various sizes; in addition many standard pieces of laboratory glass-ware constructed in *Monax* are available.

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Pyrex

Pyrex is a borosilicate glass free from alkaline earth metals, zinc and heavy metals, made by James A. Jobling & Co., and in the U.S.A. at the Corning Glass Works, where *Pyrex* was developed. The linear expansion coefficient (20–400°C) is 3.2×10^{-6} . *Pyrex* tubing is supplied in a wide range of sizes, with standard wall, heavy wall, or as extra heavy tubing. Capillary tubing and rod is supplied in a range of sizes. Precision-bore tubing with very accurate inside dimensions is also made. A very wide range of laboratory apparatus is manufactured in *Pyrex* glass.

Phoenix

Phoenix, also, is a borosilicate glass made by the British Heat Resisting Glass Co. The linear expansion coefficient (50–450°C) is 3.24×10^{-6} . Tubing and rod are supplied in a wide range of sizes. The tubing is made with either standard wall or heavy wall.

Firmasil

Laboratory glassware made from this glass is available. It is a borosilicate glass manufactured by Wood Bros. Glass Co. The linear expansion coefficient (0–400°C) is 3.2×10^{-6} .

Some Types of Glass for Sealing to Metals

Again we mention only a selection from the many sealing glasses which are made.

Lenington W.1.

This is another borosilicate glass, and was developed to make strain-free seals with tungsten. It is made by the General Electric Co., and often described as *GEC W.1.* or simply as *W.1.* The linear expansion coefficient (20–350°C) is 3.75×10^{-6} .

BTH No. 9

A borosilicate glass for sealing to tungsten, and is made by the British Thomson-Houston Co. It is often described as *BTH C.9* or as *C.9*. The linear expansion coefficient (50–470°C) is 3.65×10^{-6} .

Dial 36

Made by Plowden & Thompson Ltd. *Dial 36* is a borosilicate glass for sealing to tungsten. The linear expansion coefficient (20–300°C) is 3.6×10^{-6} .

SOME TYPES OF GLASS FOR SEALING TO METALS

Corning No. 7720 (*Nonex*)

Corning No. 7720, known commercially as *Nonex*, is a borosilicate glass containing lead and made at the Corning Glass Works. It is used for sealing to tungsten metal. The coefficient of linear expansion of *Nonex* is $3.6 \times 10^{-6}/^{\circ}\text{C}$. This is relatively low, although slightly higher than that of *Pyrex* brand glass (*Corning No. 7740*), and matches sufficiently with that of tungsten, $4.5 \times 10^{-6}/^{\circ}\text{C}$, to cause the glass to make satisfactory seals to this metal.

Nonex begins to soften, and can be worked, at temperatures a little below those required for *Pyrex*. Care must be taken when working it to avoid reducing the lead and causing the glass to blacken. The tip of an oxidizing flame must therefore be used. *Nonex* will seal well to *Pyrex*, though sometimes uranium glass is used between the two.

BTH No. 40

A borosilicate glass for sealing to alloys of the Kovar type (p. 111); it also seals to some electrical porcelain. It is made by the British Thomson-Houston Co., and is often referred to as *BTH C.40* or as *C.40*. The linear expansion coefficient is 4.85×10^{-6} from 50 to 400°C.

Lenington FCN

Made by the General Electric Co. and often called *GEC FCN* or *FCN*, this is a borosilicate glass for sealing to Kovar-type alloys. The linear expansion coefficient (20–350°C) is 4.75×10^{-6} .

Kodial

A Kovar-sealing borosilicate glass made by Plowden & Thompson Ltd, this has a linear expansion coefficient (20–300°C) of 4.9×10^{-6} .

Dial 444

This is a soda glass for sealing to platinum, and is made by Plowden & Thompson Ltd. The linear expansion coefficient (20–300°C) is 6.8×10^{-6} .

Some Types of Glass for Special Purposes

Here we mention only a few of the glasses which are made for special purposes.

Wembley L.1. Lead

Made by the General Electric Co., this glass was developed primarily for pinches and exhaust tubes in lamps and valves. It seals directly

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to platinum and to copper-clad wire (p. 111). The coefficient of linear expansion is 9.05×10^{-6} from 20 to 320°C. A lead glass, such as *L.1*, is useful as an intermediate glass for joining soda glasses of different thermal expansions.

BTH No. 12

BTH No. 12 is a lead glass for lamp and valve manufacture, made by the British Thomson-Houston Co. The linear expansion coefficient ($50\text{--}400^\circ\text{C}$) is 9.1×10^{-6} .

Wembley M.6, 'White Neutral'

This is made by the General Electric Co., and is used for medical ampoules. It loses very little alkali to aqueous solutions. The linear expansion coefficient is 7.3×10^{-6} from 20 to 350°C. This is intermediate between the soft and hard glasses, and *M.6* is sometimes used in graded seals.

Wembley 'Amber Neutral'

This glass, made by the General Electric Co., is similar to *white neutral* above, but is coloured amber by addition of iron and manganese. The linear expansion coefficient is 7.5×10^{-6} from 20 to 350°C. It is useful for ampoules when the contents need protection from ultra-violet radiation.

Leemington H.26X.

Leemington H.26X., made by the General Electric Co., is a very hard borosilicate glass of high softening temperature. The *Mg* point is 780°C. It is used in high pressure mercury vapour lamps. The linear expansion coefficient is 4.6×10^{-6} from 20 to 580°C. Sodium and potassium are absent and alumina is present in quantity in this glass.

Sodium Resistant, N.A.10

This type, made by the General Electric Co., is resistant to sodium vapour and is used in sodium vapour discharge lamps. It has a very high boric oxide content, a low softening temperature, and a low electrical resistance. It is used as an internal layer in soda glass tubing (*X.8*).

Dial 43

A borosilicate glass with an expansion coefficient intermediate between that of the tungsten-sealing and Kovar-sealing glasses. It is a useful sandwich glass for joining between these glasses. The linear expansion coefficient ($20\text{--}300^\circ\text{C}$) is 4.2×10^{-6} . It is made by Plowden & Thompson Ltd.

SOME TYPES OF GLASS FOR SPECIAL PURPOSES

Fused Silica

The Thermal Syndicate Ltd make fused silica (*Vitreosil*) tubes of several kinds. Translucent and transparent *Vitreosil* are manufactured; the former is supplied with 'sand' or 'glazed' surface, and the latter usually has a glazed surface. A glazed surface should be used in vacuum work; the transparent tubing is best. *Vitreosil* tubing is available with a wide range of sizes and with several wall thicknesses. Rods, bars and capillary tubes are available. Many items of laboratory equipment are manufactured in *Vitreosil*. They are specially valuable when high temperatures and high thermal endurance are needed. The linear expansion coefficient is 0.54×10^{-6} .

Graded Seal Glasses

The General Electric Co. makes a range of sealing glasses. For joining silica to hard glass, and hard glass to soft glass, the sealing glasses given below can be used. The coefficients of linear expansion are also given.

Sealing glass type	Expansion coefficient	Purpose
<i>WQ.31</i>	1.0×10^{-6}	for silica to hard glass
<i>WQ.34</i>	2.1×10^{-6}	
<i>H.428</i>	3.2×10^{-6}	
<i>G.S.1</i>	5.2×10^{-6}	for hard to soft glass
<i>G.S.2</i>	5.8×10^{-6}	
<i>G.S.3</i>	6.6×10^{-6}	
<i>G.S.4</i>	7.2×10^{-6}	
<i>G.S.5</i>	7.8×10^{-6}	
<i>G.S.6</i>	8.4×10^{-6}	

The G.S.1 can be joined to a hard borosilicate glass using as intermediates a Kovar-sealing glass, a sandwich glass such as *Dial 43*, and a tungsten-sealing glass.

The Ease of Working Different Types of Glass

The working of fused silica is described in Chapter 10. *Pyrex*, *Phoenix* and *Firmasil* require either a gas-oxygen flame or a gas-air-oxygen flame. All the operations we describe for *Pyrex* can also be carried out with *Phoenix*. These two glasses are very easy to work, and very suitable for research work. They join together well. *Monax* can be worked without oxygen, and is a most valuable glass for

PHYSICAL AND CHEMICAL PROPERTIES OF GLASS

laboratories, such as those in schools, where oxygen is not available. The methods we describe for *Pyrex* can be used for *Monax*. When a hand torch is used for working *Monax*, it should give a flame with a definite inner blue cone, and it may be necessary to work the *Monax* with the hand torch in small portions at a time, especially if the air pressure is low and the tubing is large. Finished articles in *Monax*, even when badly made, do not usually crack after a flame annealing. Soda glass is very easy to work in a gas-air flame, but skill is necessary to make apparatus which does not crack on cooling. Complex apparatus of soda glass cannot be made satisfactorily with a hand torch by many of the methods we describe for *Pyrex*; the bench torch methods are far better for soda glass, but they are far more difficult. We do not recommend soda glass for research apparatus unless this is of very simple construction.

The working characteristics of the other glasses mentioned can usually be deduced from their expansion coefficients. A glass of lower expansion than *Monax* needs an oxygenated flame, and one with higher expansion needs only a gas-air flame.

Distinguishing between Different Types of Glass

In a bunsen flame, soda glasses give a yellow flame and are softened, borosilicate glasses keep their hard edges, and a lead glass is blackened. When a joint is made between two glasses of different composition and pulled out, the softer glass will pull out further. The softer glass has the lower softening temperature.

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Chapter 3

GLASS-WORKING EQUIPMENT

The Glass-Blower's Table

THE usual practice is to have a special table for the bench blowpipe. This laboratory blowpipe table can be 2-3 feet square for most purposes; but for the construction of large apparatus a good size is 6 feet long and 3 feet wide. This size will permit the use of rollers for large tubes. The table should be placed so that long pieces of glass can extend beyond the ends. The top of the table should be of heat-resisting material. The blowpipe is placed near one edge, which should be flat. The other three edges should have a raised rim to prevent articles rolling off the table. Gas, oxygen and compressed air must be available; the latter is often obtained from bellows fixed under the table. Drawers for tools are desirable. A rack of some sort should be available in which hot glass objects can cool; a large block of wood bored with holes of different sizes to hold the ends of the objects is very useful for this purpose. A small flame should be present on the table for relighting the blowpipe if it is extinguished during work.

The table should be placed so that direct sunlight does not fall on the blowpipe flame. If this can happen, there should be a blind to cut out the sunlight. It is difficult to work glass in a flame in sunlight because the flame is then so difficult to see and the temperature of the glass cannot be judged properly.

The height of the table and the accompanying stool should be such that the glass-worker can rest his elbows on the table. A stool of adjustable height (like a music stool) is very convenient, and was advised by R. THRELFALL (1898). Both he and W. E. BARR and V. J. ANHORN (1949) advise a table of height 38 inches. A table of slightly less height is also satisfactory.

Blowpipes

The term 'blowpipe' is used for both the flame generator employed in glass-working and the iron tube used in blowing glass (p. 1). In the glass trade the term 'glass-blower' describes a person who uses glass melted in a pot, and the worker of tube or rod in a flame is a 'lamp-worker', who uses a blowlamp; but in the laboratory this usage is not

BLOWPIPES

customary. There a 'glass-blower' is a person who works tubing in a blowpipe flame.

Satisfactory blowpipes or bench burners are available from laboratory furnishers. They usually have a range of jets to give different flame sizes. The changing of jets during work is avoided in a burner with a turret head. The burners normally burn coal gas with either compressed air or oxygen, or air-oxygen mixtures. Crossfire burners produce a number of small flames which heat both sides of a tube at once, as shown in FIGURE 3. They are very useful for large tubing. Special burners producing a long thin flame ('ribbon burners') can be



FIGURE 3. Two crossfire burners

used for glass tube bending; they are made giving lengths of flame of 50, 100, 200 and 300 mm. A cracking-off burner giving a thin line of intense heat is also made. All these special burners can be obtained from Stone-Chance Limited.

I. C. P. SMITH (1947a) has given an account of the construction of burners for the glass-worker. The adjustment of the flame size and temperature with a given burner is best found by experience.

Hand Torches

A hand torch is a moderately light blowpipe designed to be held in the hand and moved around the apparatus. Various hand torches are available. We find the *Flamemaster* hand torch, made by Stone-Chance Limited, to be very useful. This can be fitted with a number of different jets. One gives a wide range of flame sizes with a number of gas mixtures. Three jets give different flame sizes of oxygen-coal gas flames. With the largest of these, 4-cm diameter Pyrex tubing can be worked with a single hand torch; with two hand torches as a cross-fire, 6-cm bore tubing can be worked. A double-tipping device can also be put on the torch in place of the usual nozzle; this gives two small flames at an angle to each other of about 120°. This is useful in ampoule sealing. Controls are provided on the hand torch for

GLASS-WORKING EQUIPMENT

regulating the flow of gas and air or oxygen. The *Flamemaster* can also be clamped to the bench and used as a bench torch.

Cylinder Heads and Valves

The working of glass such as *Pyrex* and *Phoenix* requires either an oxygen-coal gas flame, or an oxygen-air-coal gas flame. The oxygen is obtained from cylinders. For glass-working it is desirable to use a cylinder head which has both a regulator for controlling the pressure of the oxygen and keeping it at a steady value, and a fine adjustment valve for regulating and turning off the gas flow. In operating a blowpipe or hand torch, the oxygen is frequently turned off at the blowpipe or hand torch. The regulator then automatically stops the flow of oxygen from the cylinder. If only a valve is used the oxygen continues to come out of the cylinder, and the pressure in the rubber tubing to the blowpipe becomes so great that the tubing is forced off the blowpipe or the cylinder head; this can be very inconvenient when one is working in an awkward position with a hand torch on a complex apparatus.

Cylinder heads are liable to be damaged when they are attached to cylinders left free to roll about on the floor, and to avoid this a cylinder stand should be used.

Bellows

Foot-operated bellows fitted with a rubber disc enclosed in a string net, to provide an air reservoir, give a steady pressure of about 20 oz/in.² They are suitable for air-gas flames of moderate size. Bellows with a spring-controlled reservoir can also be used, but they do not give such a steady pressure.

Air Blowers

A compressor driven by a $\frac{1}{4}$ h.p. electric motor and giving a pressure above atmospheric of 10 lb/in.² is adequate for all normal glass-working operations, including those with a small lathe (Chapter 6). A filter should be fitted on the suction side of the compressor.

A filter pump discharging into an aspirator, fitted with an outlet at the top for air and an outlet at the bottom for water, will give enough air for small-scale operations of glass-working. W. A. SHENSTONE (1889) described this device.

Glass-Working Tools

The tools necessary for most glass-working operations are simple and can best be described in relation to their uses. Probably the most useful general-purpose tool is an iron or steel spike about $1\frac{1}{2}$ inches

GLASS-WORKING TOOLS

long set in a wooden handle (Figure 4). This spike can be used for manipulating the hot glass into the required place and in many instances can also be used in the same operations as the more specialized tools described below, although it may not be so convenient. Throughout the book this tool will be referred to as a spike.

Tools designed for various operations are marketed by a number of companies dealing in laboratory supplies. Figure 4 shows a selection of such tools. Triangular flaring tools are intended for flaring out the

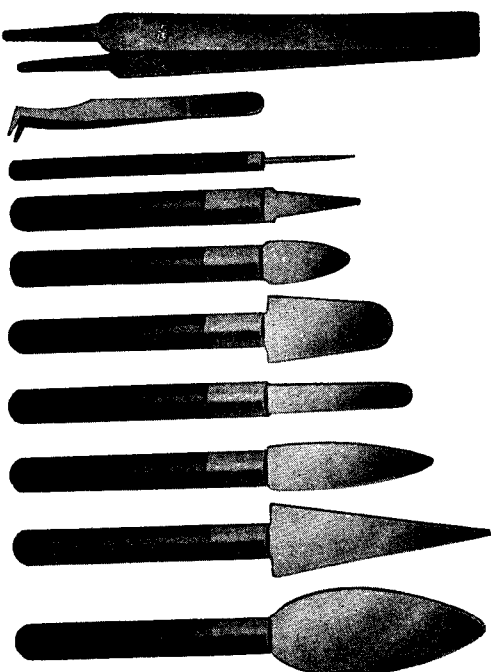


Figure 4. Some tools used in the working of glass

ends of tubes. They are usually made of brass sheet or of thin carbon plates, as are the other tools for shaping glass by means of pressure of the flat surface of the tool applied to the glass as it is rotated. The metal tools should be lubricated to prevent the glass sticking to the tool. Carbon tools require no such lubrication but achieve the same effect by wearing away in use. Hexagonal tapered reamers are used for working tubes to standard tapers for the purpose of making stopcocks or ground joints. These reamers may either be made of carbon or of metal—usually aluminium alloy. Similar tools with a much sharper taper and mounted on a handle may be found more convenient than the normal flaring tool for opening out the ends of tubes. We have found that the mounted needles from a set of dissecting instruments are useful in some cases with capillary tubing since they are fine enough to be inserted into the bore of the tube.

GLASS-WORKING EQUIPMENT

These needles have the disadvantage that, owing to their fineness, they tend to oxidize away readily and may also be melted if held in a hot flame for any appreciable length of time.

Forceps will be found useful in many circumstances. These are usually made of steel and should be designed so that there is no tendency for the soft glass to slip out easily. This will be avoided if the points are bent inwards so that they meet at an angle of about 25°. The forceps should either be long enough not to get too hot in use (about 10 inches long) or else they should be insulated in some way.

Flask clamps are made to fit a wide range of sizes of flask. These are devices for holding flasks while the necks are being worked and enable the whole flask to be rotated easily. They usually consist of a handle with three or four sprung prongs which hold the body of the flask. Some clamps are made so that they can be adjusted to fit a range of flask sizes.

Many operations require only a very modest set of tools, and too great a stress can be laid on the variety of tools required for any operation. As an example, forceps are useful for drawing off small pieces of glass or for pulling out excessively thick parts of the wall of a joint, but both these operations can be carried out by fusing a length of glass rod to the part to be drawn off and using this as a handle for the drawing operation. As in many other fields of craftsmanship, a skilful operator can work wonders with inadequate and makeshift tools, but his job would be much easier if he had all the equipment he needed, although for much of the time he may only use a very few of the tools at his disposal. There are times when the less commonly used tools will be useful, but on the other hand the glass-worker should not allow himself to be deterred by the fact that he does not have a vast array of tools.

I. C. P. SMITH (1947b) has reviewed some uses of tools in lamp-working.

Wax for Tools

Tools can be used without lubrication, but care is necessary in this case to prevent the tool oxidizing and the oxide entering the glass, which is then liable to crack on cooling. A lubricated tool slides more easily over the hot glass. Beeswax and paraffin wax have been used as lubricants; a thin film only is used. A better lubricant, advised by I. C. P. Smith (1947b), is paraffin wax mixed with some colloidal graphite. The colloidal graphite is not rapidly burnt off the tool, and it retains its lubricating properties up to 600°C. An over-heated tool can also be dipped into distilled water (5 oz) containing colloidal graphite ($\frac{1}{2}$ oz of *Aquadag*). The tool picks up a layer of graphite.

WAX FOR TOOLS

When soft asbestos is used on the bench, waxed tools are likely to pick up shreds of asbestos which can cause trouble. It is best to avoid soft asbestos in this case.

When constructing high vacuum apparatus, waxed tools should only be used when the apparatus under construction can be thoroughly cleaned afterwards. Otherwise wax may be present in the vacuum system, and will be removed by pumping only with extreme slowness.

Carbons

Carbon Plates

A few thin, rectangular plates of carbon can be very useful on the glass-working bench. The faces of these plates are of use in flattening the ends of tubing, in blowing flat-bottomed bulbs, and in shaping

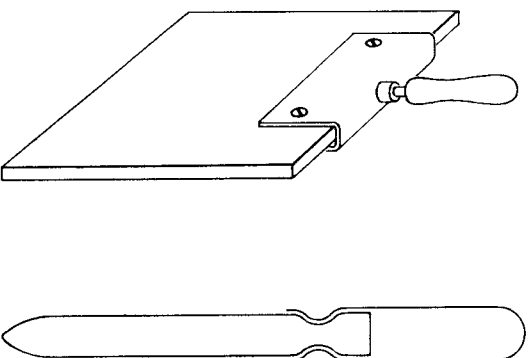


FIGURE 5. A carbon plate and a carbon rod. Both are attached to a handle

pieces of molten glass as, for example, in making the paddle for a glass stirrer. For shaping purposes it is sometimes preferable to have a handle attached (FIGURE 5). Plates ranging from $\frac{1}{4}$ to $\frac{1}{2}$ inch in thickness and of the order 3 by 6 inches are convenient sizes.

Carbon Rods

Carbon rods with one end filed to a conical shape make very good

GLASS-WORKING EQUIPMENT

flaring tools. It is useful to have a few of these handy, ranging from $\frac{1}{4}$ to $\frac{1}{2}$ inch diameter. Glass handles for these rods can easily be made by shrinking the end of a piece of glass tubing into a groove filed near to one end of the carbon rod (FIGURE 5). When working silica, carbon rods become a necessity as flaring tools, etc., because no metal will withstand the working temperature of this glass for long enough for much tooling to be done.

Gauges for Measuring

Some apparatus must be made to a definite size. The external diameter can be measured by an ordinary calliper gauge (FIGURE 6, *I*). The internal diameter can be measured either by the prongs at the

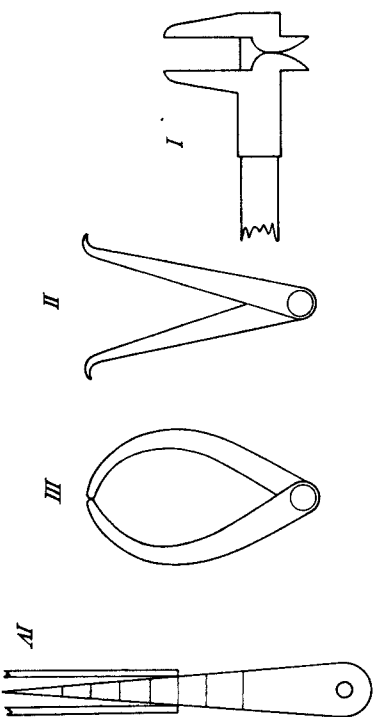


FIGURE 6. Gauges useful in measuring glass, and for working glass to fixed dimensions

back of the calliper gauge, or, for greater distances inside a tube, by an internal pair of callipers (FIGURE 6, *II*); the separation of the feet of this gauge is measured after it is withdrawn from the tube. The internal diameter at the end of a tube can be measured accurately with a cone or taper gauge (FIGURE 6, *IV*), which also reveals deviations from a circular cross-section. A pair of callipers of the type shown in FIGURE 6, *III*, is also useful in measuring the outside of a tube. The gauges in FIGURE 6, *I*, *II* and *III* can be used on hot glass; in working glass to fixed dimensions it is convenient to set a gauge at the required dimension and use it to measure the glass when it is hot.

Rubber Caps

A very convenient way of closing the ends of glass tubes for blowing, particularly with very small bore tubing into which it is difficult to fit

RUBBER CAPS

a cork or stopper, is to employ rubber caps. These are made in a variety of sizes to fit different diameter tubing, ranging up to about 30 mm, and when of good quality rubber can easily be slipped over the open end. They create a compression rather than a tensile strain in the glass tube and are therefore useful for closing tubing with ragged ends when it is desirable to avoid fire polishing. These caps are usually supplied made from rubber approximately 1 mm thick.

Thin Rubber Sheet

Pieces of very thin sheet rubber, of the order of 0.1 mm thick and of very elastic quality, as, for example, pieces cut from toy rubber balloons, can be very useful in closing ragged ends of large tubing. They are of particular use with tubing larger than 20 mm diameter. A small piece, of about the same diameter as the glass tube itself, can be stretched over a ragged end of the tube and will be found readily to stay in place by means of its own tension—provided excessive pressure is not built up when glass blowing. For detailed use of rubber caps and pieces of sheet rubber see Chapter 9.

Rubber Stoppers

Rubber stoppers are used extensively for making a closed system for blowing out. A complete overlapping range is a necessity to the glass-worker. The rubber should be of good elastic quality since poor quality stoppers often will not 'give' sufficiently to be squeezed easily into the ends of glass tubing. The stress on the glass is less and an air-tight seal is more easily obtained when good rubber is employed.

Two sets of stoppers are very useful, one set unbored and a second set with a single hole bored in them to carry blowing-tube connectors. A longer piece of glass tubing inserted into the hole will also form a very adequate handle when working small pieces of apparatus.

An assortment of small corks for closing small bore tubing can, in addition, be very useful. If the flame is to come near to a cork or a rubber stopper it is best to wrap either in thin asbestos paper before inserting into the tube. When pyrolysis of the cork or rubber stopper seems probable it is best to use a cork and not a rubber stopper, because the pyrolysis products from a cork can be cleaned off more easily than those from rubber. Sometimes a plug made from damp asbestos paper is adequate for closing a tube of small bore, and this plug can be heated quite strongly.

Blowing Tubes

Rubber tubing is necessary for blowing when the piece being worked cannot be brought to the mouth. A length of about 80 cm is convenient

GLASS-WORKING EQUIPMENT

and the size most useful is that of normal condenser tubing, or approximately 5 mm bore with 1 mm wall thickness. Lightweight rubber is advisable since heavy tubing tends to drag. It is occasionally helpful to have a finer blowing tube, and for this a length of bicycle valve tubing, about 2 mm bore, is quite satisfactory.

Mouthpieces

Mouthpieces for these blowing tubes vary with individual inclination. Some workers dispense with a mouthpiece and use the bare end of the rubber tubing. By gripping this in the teeth the flow of air can be controlled. Others desire a firmer mouth piece and use a short piece of glass tubing which must, of course, be fire polished at the end. It is perhaps also safer to strengthen the tube by thickening the end. A short piece of rubber tube, about 2 cm long, attached to the glass mouthpiece, is another modification desired by some. It combines the features of a firm mouthpiece with one where the air flow can be controlled with the teeth. A glass tube inside a rubber tube can also be used as a mouthpiece.

An assortment of glass connectors for joining blowing tubes and for connecting blowing tubes to rubber stoppers is desirable. For blocking holes in stoppers short pieces of rod or closed tubing are useful.

A holder for a blowing tube can be made out of wire. This fits round the operator's neck, and holds the mouthpiece near the mouth. It is useful when engaged in complex operations which require the use of both hands for a long time.

The Uses of Asbestos

Asbestos in one form or another finds many uses in laboratory glass-working operations. Asbestos paper or tape is very useful for blocking the ends of tubes which are so placed that they get too hot to allow the use of corks or rubber stoppers. A satisfactory blocker can be made by winding asbestos tape round a length of glass rod until it fits the tube. It may be found necessary to moisten the asbestos to make it adhere to itself. Wound round a length of tubing in a similar way, asbestos tape provides a convenient method of inserting a blowing tube into a system. Asbestos paper may be pulped with water to form a pulp which can be used for blocking irregularly shaped holes while glass-blowing is carried out on some other part of the system; but if it is used in this way care must be taken to prevent shreds of asbestos falling inside the apparatus, and one must not leave any shreds adhering to the glass when it is subsequently worked. When it is necessary to work glass under cramped conditions and

THE USES OF ASBESTOS

there is a likelihood of the flame being played on to other parts of the apparatus, it is advisable to screen these parts with sheets of asbestos board or by binding them with asbestos cord or paper. The bench is also liable to suffer from the effects of hot glass and hot tools being laid down unless it is protected by a sheet of asbestos board or similar material. Hard asbestos board is better for this purpose than the soft type as the latter has a tendency to contaminate the tools with loose fibres which may be transferred to the glass being worked.

Asbestos gloves protect the hands both from glassware which would otherwise be too hot to hold and from the effects of radiant energy. But they suffer from the disadvantage that, owing to the nature of the material, they tend to be clumsy and do not allow small objects to be picked up easily. Unless large sizes of tubing are to be worked there seems to be little advantage in the use of these gloves. But with operations with a hand torch on large Pyrex tubing, asbestos gloves are often essential—after a time the torch becomes too hot to hold, because it is heated by radiation and convection from the hot glass. The gauntlet type of asbestos glove is desirable for these operations.

When new asbestos paper and tape is used on apparatus, it may be desirable first to heat it in a gas-air flame to remove the binder.

Glass-Blowing Spectacles

It may be found that in operations in which it is necessary to observe the glass deforming or flowing whilst it is in the flame, vision is obscured by the sodium glare from the glass. Special glasses are made to filter off this yellow light and also much of the infra-red radiation. Spectacles or goggles made from such glasses are available commercially. The glass itself resembles the cobalt glass used in qualitative analysis flame tests, and may also contain rare earth oxides. Didymium goggles contain neodymium and praseodymium, and selectively absorb light in the sodium D region. The effect of the radiant heat from the glass is tiring to the eyes and may be reduced considerably by wearing ordinary sun glasses made with Crookes glass. These glasses reduce the light and heat intensity but do not significantly cut out the sodium glare. If ordinary sun glasses are used, care must be taken to see that the frames are not easily inflammable in view of the possibility of accidents.

Glass Knives and Diamonds

Glass knives are of two main kinds. The inexpensive ones are made of hardened carbon steel, and the more expensive ones are made of a very hard alloy. The latter kind keep their edge for a long time, but

GLASS-WORKING EQUIPMENT

the former soon lose their edge. Both kinds can be sharpened on a carborundum hone. For the expensive type a fine hone is desirable; for the other kind a coarse hone may be used and a scythe stone is satisfactory.

A glazier's diamond is very satisfactory for cutting sheet glass. It is not satisfactory for cutting the outside of glass tubing; the inside of glass tubing can be cut with a diamond by the device described below.

A triangular file can often be used instead of a glass knife.

Tube Cutting Device

A very useful device for cutting tubing too large to break by hand is shown in FIGURE 7.* It can very easily be made in the machine shop.

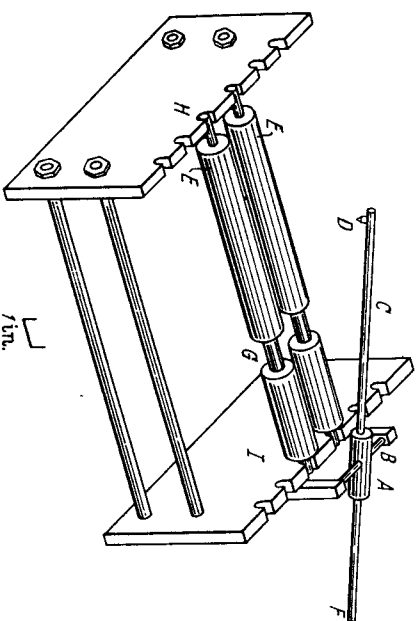


FIGURE 7. Tube cutting device

The bush *A* is hinged at *B* and carries a rod *C* whose position with respect to *B* can be adjusted. At one end of *C* is set a tiny diamond *D*. The tubing to be cut is placed over *C* so that the diamond is in contact with the inside of the glass wall and the tube is lowered on to the rollers *E*. It is then rotated through 360° while upward pressure is exerted at *F* causing the diamond to scratch the glass. A very clean break is obtained either right away, or on local heating of the glass (p. 50). The rollers are reduced in diameter over a short length at *G* to take a tube whose end is flared, as, for example, a test tube. Notches in the end-plates *H* and *I*, in which the ends of the rollers sit as shown, are cut at equal intervals on either side of the central point,

* We are grateful to Mr A. J. Hawkins, glass-blower at the University of British Columbia, whose design we have shown here.

GLASS KNIVES AND DIAMONDS

so that the distance between the rollers can be varied according to the size of the tubing to be cut.

Clamps

Good clamps for holding glass apparatus in position are essential to the scientist. Also, in many of the operations described in this book the glass-working scientist will need to hold the glass tubes in position with clamps.

The best form of clamp has both arms of the claw independently movable with a screw to tighten up each (FIGURE 8, *I*). Many standard forms have only one tightening screw, and one movable arm is

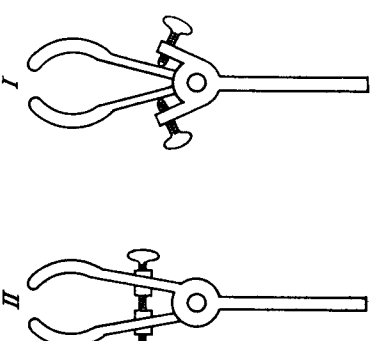


FIGURE 8. Two types of clamp

drawn towards the other whose position remains fixed (FIGURE 8, *II*). These are less convenient. Two prongs in each arm of the claw are desirable and lead to more stability than a two-and-one prong arrangement. A claw lining of asbestos or cork is preferable to rubber, and a machined boss makes for ease of manipulation and uniformity of pressure.

Abrasives

Glass may be ground to shape by using one of five abrasives. These are (1) grinding sand, (2) emery, (3) carborundum, (4) pure alumina or corundum, and (5) diamond powder. Each abrasive has its own particular property. Grinding sand is used for very rough grinding. It is mixed with water and the glass is ground to shape against a suitable surface. Grinding sand is very coarse and unsuitable for refined work, for which emery, carborundum or corundum are used. Emery is graded from coarsest to finest as 1, 0, 00, 000, 0000. The

GLASS-WORKING EQUIPMENT

grading is done by shaking the emery with water and leaving the suspension to stand. After a specified time the liquid suspension is removed and dried, so the longer the standing time the finer the emery. Emery can also be graded by particle size, by giving the size of a mesh which will just retain the particles. A 100-mesh sieve, for example, has 100 square holes per linear inch. (Often in a sieve only about 25 per cent of the area consists of holes; the remainder is the wire network. The wire size is equal to the aperture size when 25 per cent of the sieve area is holes.) When grinding glass with increasingly fine grades of emery it is essential to clean it so as to remove the coarser emery before using the finer. The glass is ground with emery in water either on a metal former or, when flat surfaces are required, on another flat piece of glass, with the emery between the two glass surfaces.

Carborundum is used mainly for fast work when much glass needs to be removed, and for grinding quartz. It is cheaper and harder than emery. It can be graded by particle size; the coarsest normally used is 90 mesh. The medium size is about 180 mesh, and the fine about 300 mesh, with very fine of 600 and 900 mesh; this last is usually unnecessary since better results can be obtained with jeweller's rouge. Carborundum is also often graded in F numbers: F consists of 240 mesh and finer; 2F of 280 and finer; and 3F of 320 mesh and finer. Carborundum is used in the same way as emery, and has the advantage that carborundum wheels and blocks can be obtained.

Corundum or alumina is again faster than emery, but it is much more expensive. Finally, diamond powder can be used, and is very fast but by far the most expensive. Once the glass has been ground it can be polished with jeweller's rouge or *Cerirouge* (see p. 83). With these polishes an optical finish may be obtained.

Glass Holders

One often needs to manipulate a piece of tubing so short that the normal method cannot be employed because the glass becomes too hot to hold. Various types of holder can be used. A rubber bung, covered with asbestos paper and bored with one hole containing a glass rod or tube, can be inserted into the tube as in Figure 9, *I*. Asbestos paper can be wrapped round a small tube until it fits into the tube to be worked (Figure 9, *II*). The smaller tube may now be manipulated with comfort and hence any desired operation can be performed on the larger tube. A variation of these methods is to employ a glass holder of the type shown in Figure 9, *III*. The holder is made by joining a small tube to a large tube; the two tubes must be coaxial. The large tubing is then cut about 2 inches from the joint.

GLASS HOLDERS

In use, the tube to be worked is wound with asbestos paper and fitted into the large tube of the holder; a rubber bung can often be used instead of the paper.

Holders for spherical glass bulbs are mentioned on p. 32. BARR and ANHORN (1949) give many examples of the use of holders of the type of Figure 9, *III*; in many laboratories, however, they are rarely or never used.

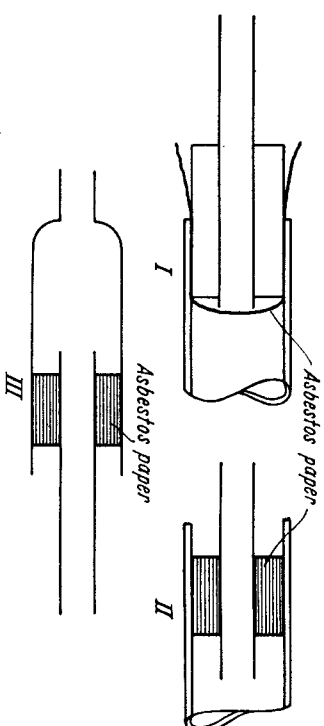


FIGURE 9. Some ways of holding glass tubes

Large Tube Supports

Large tubing is often too heavy to rotate as well as support with the left hand, so a support is used to take some of the weight. The very simple device shown in Figure 10 consists of a piece of wood with a 'V' cut in the top, and a slit to let a thumb-screw slide up and down.

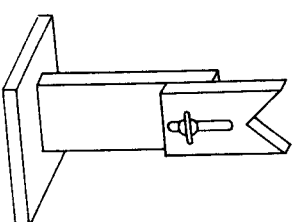


FIGURE 10. A support for large tubing

It is fixed to the stand by the thumb-screw so that the height can be varied. A few freely rotating wooden balls (e.g. small atom models) can be fixed along the 'V' to increase the ease of rotation. A clamp

GLASS-WORKING EQUIPMENT

and stand is often useful; the glass can be rested on the rod of the clamp, between it and the boss.

Adjustable Rollers

The effort required to work long heavy pieces of tubing, even with supports as described in the previous section, is tiring and may prevent adequate rotation of the glass in the flame. This effort may be reduced by the use of adjustable rollers (Figure 11), which may be in the form of two pairs of rollers on one stand or two sets of single rollers. This latter arrangement is rather more versatile, since they can be placed far apart to support long tubes, whereas the two pairs

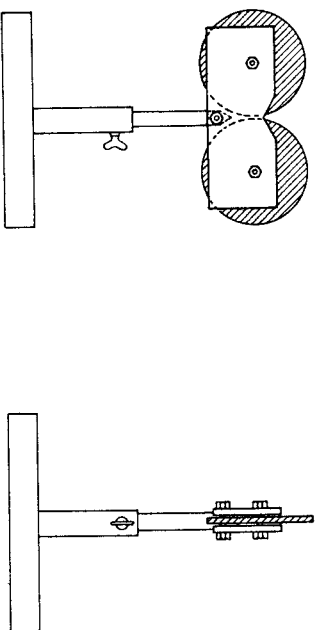


Figure 11. Adjustable rollers for rotating glass tubes

on one stand would have to be placed near the point of balance. When the glass is placed on the rollers it can be rotated easily with the palm of the left hand whilst the right hand is left free to work the glass at the end of the tube. If glass-blowing is to be carried out on the rollers, as distinct from flaring or reaming, the blowing tube must be connected by a swivel connection. These rollers can be driven by an electric motor, but for most purposes this is an unnecessarily elaborate device.

Class-Blower's Swivel

For the amateur glass-worker it is very helpful if the rotating movement of a piece of work held horizontally in the flame can be continued in a horizontal line, with the work lifted slightly out of the flame for blowing, instead of having to lift one end to the mouth. The glass-blower's swivel enables him to do this.

This small device is shown in Figure 12, *I*. It is a small union, to which a rubber blowing tube can be connected, and inside which an

GLASS-BLOWER'S SWIVEL

inner tube can rotate. It can be machined in metal or can be made quite satisfactorily from glass (Figure 12, *II*). The inner tube has holes around its perimeter directly in line with the union T-piece. If made from glass these holes can be formed by puncturing molten glass with a tungsten needle, or by drawing out tiny spindles and breaking these off. The inner tube must fit closely into the tube in which it rotates, to minimize air leakage.

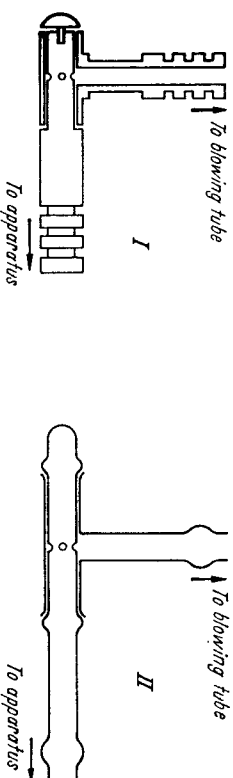


Figure 12. Two designs of glass-blower's swivel. *I* is of metal and *II* is of glass

Polarized Light Strain-Viewer

When intricate pieces of apparatus have been constructed it can be useful to observe the presence or absence of strain with a polarized light viewer. A. L. REIMANN (1952) gives descriptions of various systems and explains the optics of these systems.

The most sensitive form has a diffused white light shining through a polaroid screen, then a mica tint plate and a second polaroid screen crossed with respect to the first. The piece of glass apparatus under examination is held between the first polaroid screen and the tint plate. Unstrained glass appears coloured violet-red, and the presence of strain is indicated by change of colour in the strained portions through red or blue-green to yellow, depending on the nature and amount of strain. The orientation of the work must be varied to give maximum colour change, otherwise quite a large strain can be missed. This can happen because glass subject to simple axial tension or compression behaves as a uniaxial crystal with optic axis along the direction of stress. Along this direction, therefore, no double refraction occurs.

The tint plate produces a retardation of about 570 millimicrons, so that the yellow components of white light vanish. The resulting colour of violet-red (or 'sensitive violet') is changed when an extra retardation is introduced by the strained glass. The eye is very

GLASS-WORKING EQUIPMENT

sensitive to colour changes in this region. J. H. PARTIDGE (1949) has given a table of colours observed in a strain-viewer without and with a tint plate.

Glass-Cutting Wheels

Large glass-working laboratories may be equipped with a glass-cutting machine. This usually consists of a high-speed, power-driven, fine abrasive wheel, or, better, a steel wheel in whose perimeter are embedded fine diamond particles. A lubricant—water, or water and cutting oil emulsion—is played on to the faces of the wheel from jets on either side. The glass to be cut is held on a movable steel table mounted on rollers.

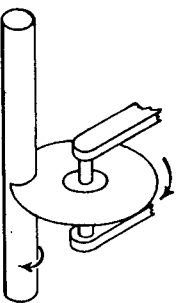


FIGURE 13. Cutting through a glass tube with a high speed wheel

For an oblique cut the glass can be clamped in position. When a straight cut across the length of the tubing is required the tube is carefully rotated during the last stages of the cut, in the direction shown by the arrow in FIGURE 13; this keeps the wall of the tubing, at the point of cutting, perpendicular to the cutting blade and prevents a portion of the wall from breaking away at the completion of the cut. By varying the manipulations the ends of tubing can be slotted and small holes or slots can be cut in the walls. A small hole, for example, is produced when the cutting operation is stopped just as the wheel breaks through the wall.

Lapping Wheel

For many grinding operations a lapping wheel can be most useful. A horizontal metal table rotating at about 30 r.p.m., sprinkled with carborundum powder and moistened with water or cutting oil-water emulsion, forms a very satisfactory grinding table. A raised rim (see FIGURE 14) prevents the grinding paste from being thrown from the circumference of the table while an inclined baffle, as shown,

LAPPING WHEEL

made of wood with a lower strip of leather, keeps the paste from piling up on the outside.

In most cases the glass being ground can be held by hand, but for making accurately parallel faces, for instance, some form of adjustable mounting is useful. This can usually be built to suit the particular operation.

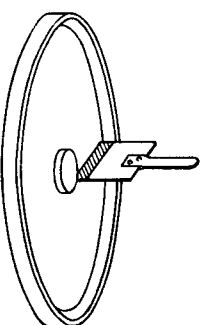


FIGURE 14. A lapping wheel

Carborundum Grinding Wheel

A very fine, high-speed, carborundum wheel—of the sort employed for grinding very hard metals—can be used for grinding glass. A delicate pressure must be used, particularly with tubing, to avoid shattering the glass. The wheel must be continuously moistened either with a jet of water or by holding a wet cloth against a portion of its face.

The method gives a very coarse finish but enables a large piece of glass to be ground down quickly. The work must be finished with carborundum powder by hand or using a lapping wheel.

Annealing Oven

Glass-working shops that are called upon to make large or intricate apparatus should be equipped with an annealing oven. With some ingenuity a home-made one can be built using *Nichrome* wire and fire-brick.

The glassware should be transferred to the oven as soon as possible and the temperature slowly raised to the annealing point of the glass used. This temperature must be held for from five to ten minutes or more and the oven then allowed to cool slowly. The maximum cooling rate depends on the glass and wall thickness, and data can usually be obtained from the manufacturer. For *Pyrex* brand glass, *Corning No. 7740*, and wall thickness not greater than 2 mm, the maximum cooling rate allowed should be 56°C/min. For a wall thickness of 5 mm the cooling rate should not exceed 9°C/min. Annealing schedules for some glasses are given on p. 20.

GLASS-WORKING EQUIPMENT

Sometimes it is possible to transfer the article straight into a hot oven. This depends upon the particular piece of apparatus. With a diffusion pump, for instance, where most of the joints have been allowed to cool completely before the final parts are assembled, the whole article is placed in a cool oven which must slowly be raised to the annealing temperature. On the other hand, it may be desirable to transfer a complicated part, a multiple glass-to-metal pinch seal, for example, directly from the glass-working flame to a hot oven already at, or near to, the annealing temperature. This is the case when cooling without careful annealing can set up large strains.

Treatment of Slight Burns

Trivial burns are sometimes produced by the accidental handling of hot glass. These may be smeared with vaseline or skin cream and covered with clean or preferably sterile linen. Blisters should not be broken. Tannic acid jelly is not recommended.

Amount of Equipment required for Different Purposes

The various items of equipment we have described are all useful in a laboratory where a great deal of difficult glass-working is carried out. Much difficult work can be done without a tube-cutting device, glass holders, large tube supports, adjustable rollers, a glass-blower's swivel, strain viewer, cutting wheel, lapping wheel, grinding wheel and an annealing oven. These items, however, together with a glass-working machine (Chapter 6), are desirable in a laboratory where all kinds of difficult jobs are tackled. In school laboratories only simple jobs will normally be done. The necessary equipment for such jobs consists only of a hand torch, an air blower or bellows of some kind, a glass knife, a spike, a flaring tool and forceps, and a set of rubber stoppers with a blowing tube. Stands and clamps will be available as ordinary laboratory equipment. An asbestos board is useful to protect an ordinary bench used for glass-working.

In research work a lack of the proper facilities may lead to a considerable loss of time because jobs are done far more slowly when the proper equipment is not available. For example, much time is lost when a complex apparatus cracks because an annealing oven is not available, and when large tubing is manipulated without a glass-working machine. We think that R. THRELFALL's (1898) remarks are still worthy of note. He wrote 'Though, no doubt, a great deal can be done with inferior appliances where great economy of money and none of time is an object, the writer has long felt very strongly that English physical laboratory practice has gone too far in the direction of starving the workshop, and he does not wish, even indirectly, to

AMOUNT OF EQUIPMENT REQUIRED
give any countenance to such a mistaken policy. Physical research is too difficult in itself, and students' time is too valuable for it to be remunerative to work with insufficient appliances.'

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Chapter 4

BASIC GLASS-WORKING OPERATIONS

The Preliminary Preparation of Materials

BEFORE any glass-blowing is started it is highly desirable that the tools and all pieces of glass rod, tubing, etc., should be collected together and laid out systematically. At times this may seem unnecessary and a waste of time, but it will be found to be worth-while in the long run. When deciding what will be required, the complexity of the job in hand must be borne in mind: the more complex the job the more care required in preparing the materials and in working out mentally the way in which it is to be done. If this is not done, sooner or later it will be found that some tool is missing when it is required and that the work cools and cracks while the tool is being sought. B. D. BOLAS (1921) wrote 'A large amount of glass-blowing is spoiled through carelessness in arranging the work beforehand'.

The need for system and order in all practical work has been stressed by M. FARADAY (1842), who wrote, in giving general rules for young experimenters, 'They all relate to *method*, that great source of facility and readiness which is equally influential in the performance of the most common and the most difficult processes.' His remarks are as pertinent today as when they were made, particularly as the tradition of the handing on of practical knowledge is not as strongly maintained as it was in the latter part of the nineteenth century. (R. W. Bunsen used to teach his students glass-blowing and other practical arts and even went so far as to make the more difficult pieces of apparatus himself.)

There seems to be a general psychological tendency in doing practical work with glass (and other things) to try and take short cuts and save time by using improper and hazardous methods, and in our experience this tendency in the end leads to more waste than saving of time. One must also avoid the temptation, in any given case, to think that just for once a risk can be taken.

The Cleaning of Glass Tubes

Before any glass-working is started care should be taken to make sure that the glass is thoroughly cleaned. This cleaning is necessary for

THE CLEANING OF GLASS TUBES

two main reasons. One is that if inorganic material is left on the glass it will not burn off in the flame and will become embedded in the glass. This can, under some circumstances, lead to local strain in the joint and may also prevent the two pieces of glass fusing together properly, leaving a pinhole in the joint. The second reason why thorough cleaning is important is that it is essential, if the apparatus is to be used for high vacuum work, for the inner surfaces to be grease-free. It is much easier to do this cleaning before a complex piece of apparatus is assembled.

Normal chemical cleaning procedures are satisfactory for cleaning glass tubing for glass-blowing. Much of the dirt can be removed by the use of water, a detergent and a long brush such as a burette brush. It is desirable not to scratch the inside of the tube with the wire of the brush. The use of a brush can be avoided by pushing or pulling a piece of soft cloth through the tube. Chromic acid is satisfactory for cleaning glass, after removal of the gross dirt, provided care is taken to see that the glass is free from mercury. The usual laboratory cleaning solution, known colloquially as 'chromic acid' contains about 50 ml of saturated aqueous sodium dichromate in a litre of concentrated sulphuric acid. If mercury is present a residue is precipitated which is difficult to remove. Thorough washing with water is necessary after cleaning in chromic acid. Organic solvents are adequate for removing grease from the inner surfaces of tubes where no other dirt is present; while acetone or alcohol may be useful for drying glassware, care should be taken that liquid or vapour from these solvents does not remain inside the glass when glass-blowing is commenced, because of the risk of explosion. When glass which has been in contact with mercury is worked, great care should be taken to clean the mercury from all parts of the system likely to become hot, since mercury vapour is toxic.

It is very difficult to remove the last traces of chromium from glass cleaned with the usual chromic acid mixture. The remaining traces may cause error in oxidation studies. R. A. WILKINSON (1955) found serious errors from this cause in fat oxidation studies, and he cleaned greasy pipettes in a boiling, $\frac{1}{2}$ per cent aqueous solution of "Teepol" (sodium secondary alkyl sulphate). Glass can also be cleaned in boiling aqua regia.

Breaking Glass Tubes

Tubing up to about 15 mm diameter can be broken by scratching the surface in one place with a file or glass knife, making the scratch at right angles to the axis of the tube and then placing the thumbs on the side of the tubing opposite to the scratch; the thumbs are pressed

BASIC GLASS-WORKING OPERATIONS

forward and the rest of the hands are slightly pulled apart (FIGURE 15). This is a gentle process and not a test of strength. The scratch on the tubing can, with practice, be made with one stroke of the glass knife; it is not necessary to try and cut through the wall of the tubing, and the knife is soon blunted when it is used like a saw. Most glass-workers have the impression that wetting the initial cut makes the glass break more easily. Soda glass is easier to scratch with the knife than the borosilicate glasses.

We find that tubing in the range of 15–30 mm diameter (approximately) can be broken by a variation of the above method: The tubing is scratched fairly strongly, and placed, with the scratch uppermost, so that it rests on a copper wire, about 3 mm diameter, which is opposite to the scratch and parallel to it. The scratch is moistened,

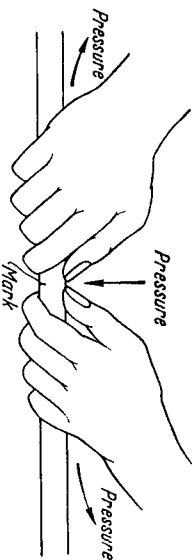


FIGURE 15. Breaking a small glass tube

and the ends of the glass tube are forced down strongly while the tube is balanced on the copper wire: a square break is then usually obtained. This method works best with moderately thin-walled tubing.

Tubes exceeding 15 mm diameter (and smaller) can be broken by the 'hot-spot' technique: A fairly deep scratch is made in the glass in one place, and wetted. The point of a piece of glass rod is then heated white hot, to give a hot spot a few mm in diameter, and this is placed on the end of the scratch. The glass may crack right away, or it may crack when the hot spot is removed and damp cotton wool is applied to the hot glass. Usually, with borosilicate glass, the crack goes only part of the way round the tube; a soda tube often cracks completely. The crack can be continued by placing another hot-spot against its end. This is continued until, with luck, the ends of the crack meet. With borosilicate glass the crack usually wanders, and a square end is not obtained. A cracking-off burner (p. 29) can be used in place of a hot spot.

A square end is more easily obtained by the hot wire method. The tubing is first scratched all the way round. This may be done by

BREAKING GLASS TUBES

hand, a line being first drawn round the glass with a wax pencil—the line can be checked to see that it is perpendicular to the tube axis—and the scratch made along the pencil mark. Without the pencil mark, the scratch may deviate from the perpendicular. A peripheral outside scratch may also be made by rotating the tube against a rotating steel wheel. An internal scratch may be made with a diamond by the method described on p. 38.

After scratching, 26 s.w.g. (or thicker) *Nichrome* wire is wrapped once round the scratch, which is wetted. The wire is heated to redness electrically, and kept taut round the glass—for this purpose it is convenient to have insulated handles on the wire. The tubing may then crack smoothly; if not, it may be induced to crack by touching the scratch with wetted asbestos when the wire has been hot for about half a minute. An internally scratched *Pyrex* tube may not crack until it is put in a flame.

Sometimes a large bottle is cut to make a heavy beaker or a bell jar. Big bottles are generally of soda glass, and can be cracked with a flame. The bottle is first fairly heavily scratched all the way round, and then a very small flame is directed on to the scratch. A crack should start, and it can be led round by the flame. A cracking-off burner (p. 29) is specially useful here. This operation is quite difficult for beginners, because the glass often just shatters; but with practice it can be mastered. Increased safety is obtained by wrapping strips of wet blotting paper or asbestos paper round both sides of the crack, leaving a gap, about $\frac{1}{2}$ inch wide, containing the crack. An alternative method for large bottles, etc., is to spin them on a turn-table and to have a fine pin-point flame playing at the level of a small initial scratch; after a few minutes the flame is removed and a piece of wet cotton wool pressed on the scratch.

I. C. P. SMITH (1947) has described a simple method of grinding through glass tubing with a hacksaw: the blade is replaced by soft wire (such as iron, copper, or *Nichrome*) of 24 to 18 gauge, and this is ground through the glass, using 180-mesh carborundum powder and water and an ordinary sawing action. A glass-cutting wheel is described on p. 44.

Knocking off Small Bits of Tubing

Sometimes there is not enough tubing left beyond the scratch to hold in the hand, so that the normal method for small tubing (p. 49) cannot be used. The easiest method to use in this case is to scratch the tube and then place it on the top of a triangular file, with the scratch opposite to and above the edge of the file (FIGURE 16). The end of the tube is then knocked off by giving it a sharp blow with a heavy object.

BASIC GLASS-WORKING OPERATIONS

Often the end of a broken tube is not quite square. This can be somewhat corrected, especially in larger tubes, by using a piece of wire gauze: The gauze is stroked against the end of the tubing, when small pieces of glass are broken off the protruding edges until the end assumes a roughly square shape. Another method is to heat the end of the tube strongly and cut off the protruding parts with scissors.

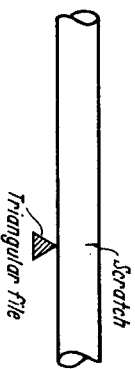


Figure 16. Knocking the end off a small piece of tubing

Holding and Rotating Tubes in the Bench Flame

Holding and rotating tubing in the flame is a most important part of glass-blowing with small tubing. Usually the weight of the glass is supported by the left hand, which is held over the tubing, and the right hand, under the tubing, maintains only a light grip. Although this may feel odd to the beginner, it will be found satisfactory because the right hand is free to do the manipulations and to bring the tube to the mouth for blowing. However, the right hand may also hold the weight with an overhand grip while the left hand has the underhand grip. In research work it is convenient to be able to use both methods.

The more usual disposition of the hands is shown in FIGURE 17.

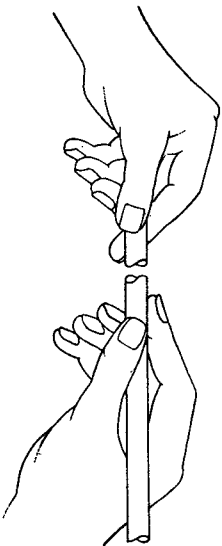


Figure 17. Holding and rotating a tube. The disposition of the hands is as seen by the operator

The tubing is rotated by the forefinger and thumb of the left hand, and rotates in the groove between the thumb and forefinger of the right hand, and can rest lightly on the middle finger of the right hand. It is easy enough to get a smooth rotation of the glass when it is rigid, but when it is soft both hands must rotate the glass without distorting

HOLDING AND ROTATING TUBES IN BENCH FLAME

It, and this is difficult (p. 2). One must get the 'feel' of the glass. Its softening is a gradual process, and its 'feel' during working is determined by viscosity, surface tension, thickness etc. Practice is necessary here.

The glass should be rotated horizontally, unless it is desirable for it to flow towards one end. Usually it is rotated with the upper side moving away from the operator, but this seems a personal preference. When the glass becomes very hot and plastic, it is essential for the right hand to keep the tubing rotating at the same speed as the left-hand piece, or streaks and ridges will appear. This operation is found difficult by some people, but others master it with practice.

With soda glass the rotation of the article in the flame is of prime importance, since if part of the work is allowed to cool, re-heating may cause cracking and the destruction of the article; but with Pyrex and similar glass it is often possible to heat only a portion of the article at a time.

When the tubing is too heavy to hold and rotate, one end can be rested on a support (p. 41), or rollers can be used (p. 42).

The beginner should not be disappointed if at first these operations produce a hopeless mess; with practice the facility is gained.

Bends and Spirals

To bend a piece of tubing, the left-hand end is first closed. The tube is rotated in a fairly big flame with not very much air (for soda) or oxygen (for Pyrex) in it. A special flame from a ribbon burner (p. 29) is best, but not essential; an ordinary bench blowpipe gives an adequate flame. The tube is held as in FIGURE 17. When soft over the desired length it is brought, held horizontally, up to the chest. The right hand then brings its end vertical, and gentle blowing into this end brings the tube into shape with no kinks or flat portions. If a bad bend is made it can be corrected by heating and blowing. Many other methods of making a bend have been described.

Spirals

To make spirals of uniform size throughout is difficult; but useful spirals can be made fairly easily by a beginner even though the product may not look too elegant: Longer sections than usual must be heated, and care must be taken not to flatten the tubing. It is easier to make spirals with smaller tubing because the ratio of wall thickness to diameter is highest.

Spirals can be made either free-hand or with a mandrel or former. For the free-hand method a right-angle bend is made (FIGURE 18, A), with a length of tube (between A and B) of about 6 inches left to act

BASIC GLASS-WORKING OPERATIONS

as a handle. The tube is next bent into a right angle at C (FIGURE 18, *II*) so that the length BC is equal to the desired radius of the spiral. It is then heated near C, both by rotation and lengthways, over a fairly large area—the major part of the heating being on the outside. When the glass is soft enough, the length AB is held horizontally in the left hand and the length CD is slowly bent to form the first arc of the spiral (FIGURE 18, *III*). The tube is next heated adjacent to the first arc, and, when sufficiently soft, is bent to continue the arc previously made. The bending is continued until a complete turn is made. The next step is to ensure that the tubing AB is axial. This is

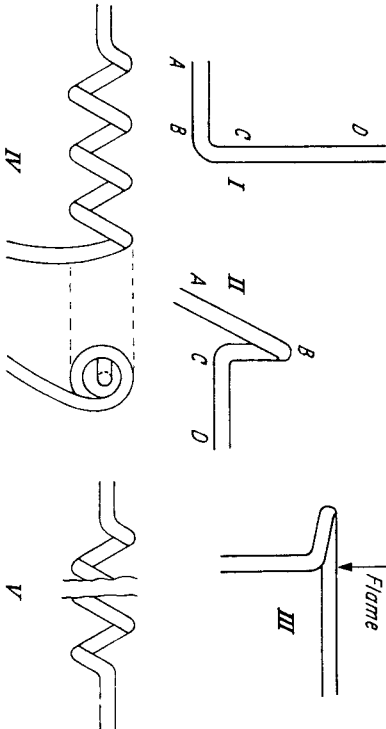


FIGURE 18. The production of a spiral

done by heating at B and bending AB into the correct position. All the bending should be done outside the flame and it is advantageous to look at the spiral from various angles to check the uniformity of the coils. The spiral is completed by continuing as before.

To make an elegant-looking product it is necessary to pay attention to the uniform diameter of the coils and to keep the spacing between them constant. The final product (FIGURE 18, *IV*) usually has to have the two ends aligned. To do this, the last turn is heated at a distance from the end of the last arc approximately equal to the spiral radius, and a right-angle bend is made. The end of the last arc is then heated, and this right angle is pushed inwards. Final adjustments can be made after rotating the spiral to check alignment. The spiral should appear as in FIGURE 18, *V*.

It is often easier to wind a spiral on a mandrel or former of carbon, metal tubing or glass tubing wrapped with asbestos paper. A right-angle bend is made in the tube to give a short portion at the end which

BENDS AND SPIRALS

is strapped to the mandrel. The tubing is heated in sections of about 1 inch and wrapped round the mandrel. The asbestos can be used as a guide to ensure that even turns are made—one way is to wrap

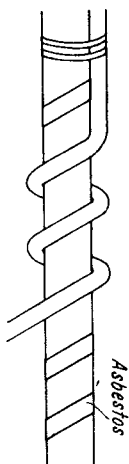


FIGURE 19. Production of a spiral with a mandrel

asbestos tape round the mandrel in a spiral path (FIGURE 19). Finally the asbestos is softened in water and the spiral is removed from the former.

A flat circular spiral can be made by the free-hand method: a right-angle bend is made, and the glass near the bend is bent round the short section used as a handle. By repeated heating and bending a flat coil is produced.

Putting a Handle on the End of a Tube

When it is necessary to work comparatively near to the end of a piece of tubing it usually becomes too hot to hold. It may be inconvenient to support the whole weight of the glass in one hand, and the unsupported end will almost certainly drop under its weight when the tube is hot. A handle on the end of the tube is therefore useful.

First let us consider the case of a tube which is to be drawn off near the end. In this case the end need not be airtight. The tube to have the handle attached is held in one hand and the piece of rod or tubing to be used as a handle in the other. The end of the tube is rotated in the flame until the glass is soft for a length about equal to its radius. The handle should be heated until its end is molten and then attached to a selected point on the end of the tube. The tube should be taken out of the flame and the handle pulled across until the point at which it is attached to the tube comes into contact with the far side of the tube. This should be heated so that the two sides of the tube and the handle are thoroughly fused together and the handle should then be pulled approximately central on to the axis of the tube. The whole mass of glass should be kept heated for a short while by rotating in the flame to allow the strain in the glass to be reduced. Such a handle should not crack off when the glass cools, and is very useful when the tube has to be drawn off or bent near to the end. It does, however, leave the end of the tube in such a state that it has to be either drawn off or cut off and no blowing can be carried out as it is not air-tight.

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A handle can also be fused on to the closed end of a tube to support it while glass-blowing operations are carried out: This is done by heating the end of a piece of rod for the handle, heating the centre of the closed end of the tube, and fusing them together. When the handle is to be removed after the operations are finished the point where it is fused on can be heated, the handle drawn off, and the end of the tube blown out again to its original shape.

With big tube it may be convenient to prepare a handle from rod shaped at its end like a Y, so that it sticks to two points along a diameter at the end of the tube. It is then not necessary to push in the ends of the glass tube. The Y-shaped handle is specially useful for very big tubes.

Drawing Tubes off to Spindles

Drawing a tube off to a spindle is the simplest of the basic glass-working operations. It is a convenient alternative to cutting glass tubing when short lengths are required and it is the preliminary stage to many other operations.

To draw off a tube using the flame of a bench torch, one places it in the flame and rotates it, so that heating is uniform, until the tube starts to collapse under the surface tension of the molten glass. The glass should then be removed from the flame and the two parts drawn steadily apart at a rate which gives a strong spindle as the glass cools. The spindle can then be cut by melting it in the flame and pulling the two bits apart. A more axial spindle is obtained when rotation is continued as the glass parts are drawn apart.

It may be convenient (specially for big tubes) to shape the glass somewhat in the flame before drawing the spindle. When the glass starts to soften, the two parts of the tube can be drawn apart slightly to maintain an approximately uniform thickness of glass as the tube is drawn down. In this way a tube may be tapered or constricted. If the two parts of the tube are drawn apart too rapidly, before the glass has collapsed sufficiently, thin walls are obtained, and, if they are not drawn apart enough, thick walls result. The rotation of the two parts of the tube must be the same and the tube must be kept approximately straight during this process if a symmetrical taper or constriction is to be obtained. Continuation of the process greatly reduces the size of the tube at the point of heating.

The spindle makes a very useful handle for holding the tube while other glass-blowing is carried out, and owing to its much reduced diameter it does not become hot when the tubing is worked near to the end. The tip of the spindle can be cut off to attach a rubber tube for blowing. If for any reason an accurately central spindle is

DRAWING TUBES OFF TO SPINDLES

required it may be found convenient to put in a second spindle using the first one as a handle, since short lengths of tubing are normally more easily worked than long pieces which are unevenly balanced about the point of working. The operations involved in putting a central spindle near the end of a piece of tubing are shown in FIGURE 20. In many glass-blowing operations, a long spindle is extremely useful. Spindles can be closed very quickly in a flame, or opened again with a glass knife; we find it preferable to use them, when possible, instead of rubber stoppers.

Experience of the feel of the molten glass as the tube is rotated and

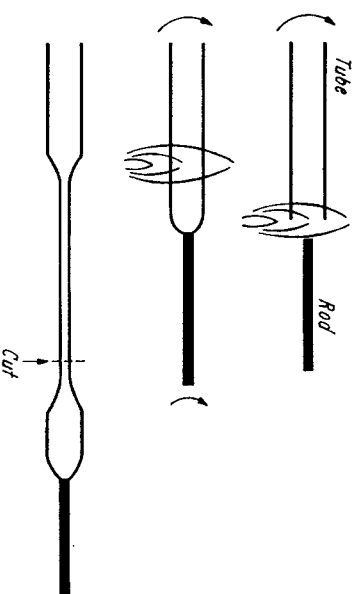


FIGURE 20. Drawing tubes off to spindles

drawn down will enable a happy medium, between the extremes of thin-walled spindles and spindles of glass rod, to be obtained.

The operation of putting a spindle in a tube is often described as 'pulling a point' or 'pulling a spear'. This is a most important operation in bench flame work. The spindle must be central, otherwise the rest of the work will not be central.

Round-Bottomed Closures of Tubes

When a good spindle is obtained and a rounded end is required the glass should be heated at the end of the taper (as shown in FIGURE 21): the glass here should be collapsed down in a small flame and pulled out comparatively slowly so as not to leave a blob of glass on the end of the tube. The whole end of the tube should then be heated by rotating it in a fairly large flame until the glass at the end of the tube has started to collapse a little. The tube should be removed from the flame at this stage and should be blown into gently. The glass should be heated relatively strongly—most beginners tend to try to

BASIC GLASS-WORKING OPERATIONS

work the glass before it is hot enough which leads to an uneven finish. It is advisable in the first place to blow gently so that the end of the tube is stretched slowly. In this way the blowing can be stopped at any required stage, and this is desirable since if any part of the glass is thinner than the rest this part will start to blow out first because of its lower viscous drag but will cool faster than the thicker parts which will continue to blow out after the thinner glass has become too cool to extend much more. In this way a more even rounded end will be produced than if the glass had been blown rapidly and then allowed to cool.

The most usual fault in making a rounded end is to leave a lens of thicker glass at the point where the spindle was drawn off. This may be because either insufficient glass was drawn away with the spindle—or if this is the case more can be pulled off with a piece of rod—or else

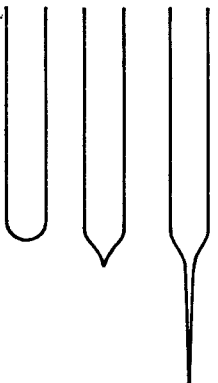


FIGURE 21. Closing a tube with a round end

the glass was not heated enough to allow the unevenness to even itself out by flowing under surface tension. A lens of thicker glass left in a round-bottomed closure can be removed by heating it with a small flame and blowing it into a small bulb. The whole end of the tube should be re-heated by rotation in a medium-sized flame until the glass starts to flow. The end is then blown out to a hemispherical shape and annealed.

Fabrication of Thin Glass Rod

In some of the following sections the use of thin 'spills' of glass rod (welding rod) is described. They are most easily made from glass rod by heating a portion and pushing the ends together until a reasonable amount of molten glass is accumulated. This is drawn out at a rate which gives a rod of 1–2 mm diameter.

If no rod is available the thin glass rod can be made from tubing: thick-walled capillary tubing is the most convenient but normal 8 mm tubing will do quite well. A length of tubing, about 10–15 cm for

FABRICATION OF THIN GLASS ROD

thick-walled capillary or 15–20 cm for 8 mm tubing, is rotated in the flame until a part collapses down and fuses to a solid mass. This process is continued, pushing the ends inwards slightly to build up a mass of fused glass with no fine capillary through the centre. Such a capillary is easily seen while the glass is hot. The molten glass should be drawn down in the same way as for rod, taking care that the tube next to the fused mass does not draw down to form a length of fine capillary. This will be prevented if the initial stage of the drawing down is carried out very slowly. Straight rod is obtained by rotating during the drawing-out.

Blowing Holes in Glass Tubes

There are basically two sites for blowing holes in glass tubing, the first where the hole is to be made in the end of a tube which has been drawn off, and the second where the hole is required in the side of a tube. The principle is the same in both cases: the glass to be blown out is heated until it is soft enough to flow easily and is then blown out into a very thin bubble. The bubble is then removed. If the glass is too hot or it is blown too hard, the bubble will not be formed, but a small irregular hole will be blown and a cloud of very fine glass flakes produced. This is undesirable.

In the case in which the end of the tube is to be blown out care should be taken to see that the tube is heated evenly over the whole end, otherwise the hole produced in it will be skew. If the hole is to be smaller than the bore of the tube only a small area of the rounded end must be heated and the blowing must not be continued too long, since once the bubble of glass is formed it is enlarged principally by drawing more glass from around the edge of the hole in the original tube, thus making the hole larger. This also applies to the blowing of holes in the side of a tube.

The second case, the blowing of a hole in the side of a tube, is a necessary preliminary to making a T-joint, and therefore it is essential to be able to produce the right-sized hole in the correct place. The procedure is very similar to that for blowing out the end of a tube, except that a small area on the side of the tube is blown out. Generally speaking it is more convenient to heat a comparatively large area on the side of the tube and blow a slight bulge, and then heat the smaller area to be blown out more strongly with a rather smaller flame. This should give a hole, if blown out correctly and not for too long or too hard, which is surrounded by shoulders so that the rim of the hole stands clear of the general line of the tube walls. In many cases it is desirable to have these shoulders present (e.g. making T-joints).

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The removal of the thin glass bubble from the hole does not present any problems. It can be easily knocked off, but this scatters flakes of glass which may get inside the glass being worked. The glass bubble may be crushed in a sheet of paper, which certainly causes less mess and may be preferred in the interest of general tidiness when the work is being carried out on fixed apparatus. A solution to the problem, which does not cause flakes of glass and leaves a supply of glass for further working about the hole, is to fuse the bubble down into a rim around the hole. This usually happens as a matter of course when a fairly cool flame is played on the thin glass, but if too hot a flame is used the glass is boiled which does not improve its properties (see Devitrification, p. 18).

To blow a very small hole the method described for multiple perforations in bulbs is useful (p. 72).

Joining Tubes of Similar Size

In this section detailed descriptions of the operations will be given for the benefit of the complete novice. It is convenient to deal with the subject in three sections, (1) the case in which both tubes can be rotated in the flame of a bench torch, (2) the case of one fixed tube and the other movable, and (3) the case of two fixed tubes.

(1) *Joining Two Tubes in the Flame of a Bench Torch*

The first step in producing a satisfactory joint is the preparation of the two ends to be joined. The most satisfactory method is to draw the tubes off and to blow the ends out. If this is done correctly the ends should be square. Cutting the tube with a file scratch leaves a line of fine bubbles in the joint due to the remains of the original scratch. This spoils the appearance of the joint when finished and may also cause a weak spot at which a pinhole may develop when testing for leaks with a Tesla coil. Jagged ends of tubing can be heated until the glass is quite soft and trimmed with scissors.

The second step is to block the end of one of the tubes and heat the ends of the two tubes in the flame until the glass is quite soft and starts to collapse. Uniform rotation is necessary for this, as in all the heating operations for this method. When this stage is reached the glass is removed from the flame and the ends pressed lightly together; initial contact is best made at some point by holding the two tubes at a slight angle (FIGURE 22). The ends are then brought together by bringing the two pieces of the tube into line. A slight pressure is then applied during the heating of the join so that the glass is thickened around it. This glass is worked thoroughly by alternate heating and blowing, care being taken to see that the two

JOINING TUBES OF SIMILAR SIZE

parts of the tube are rotated at the same speed during the heating process. When the glass has been worked sufficiently, so that the original junction of the two tubes cannot be distinguished, a slight bulge is blown in the tube and it is drawn down until the walls are of uniform thickness and the tube has a uniform bore. The whole operation is sometimes described as 'butt sealing' or 'butt welding' and the joint as a 'butt seal'.

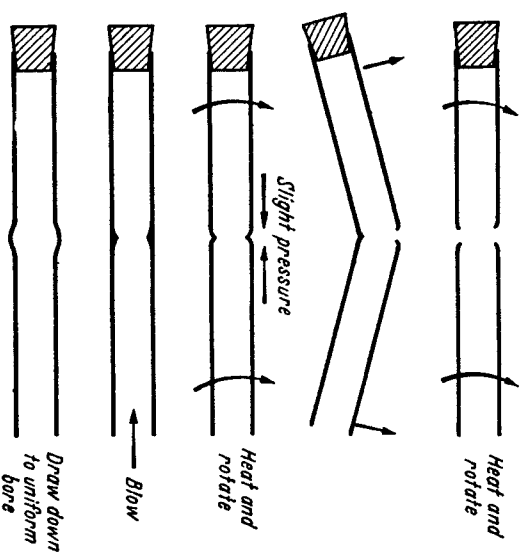


FIGURE 22. Joining two tubes

It will probably be found that much practice is necessary before good joints can be made in this way. A common difficulty is that the two halves of the tube must be rotated at the same speed, and to avoid this many novices tend to work round the joint in segments once the two pieces of tube have been pressed together. This method, while capable of producing vacuum-tight joints, leaves an irregular wall thickness which is much more liable to crack from strains induced on cooling, even though it may have been carefully annealed by cooling slowly in the flame. However, it is satisfactory for Pyrex and similar glass.

(2) *Joining Two Tubes, one of which is Fixed*

The necessity of joining a tube to another which is fixed arises frequently in the construction of laboratory apparatus where the

BASIC GLASS-WORKING OPERATIONS

main part of the apparatus is either fixed to a stand or too large and unwieldy to allow the use of a bench torch. It is very often easier for a novice to produce a good joint by clamping the glass in position and working with a hand torch as described below, since it does not require the same degree of manual dexterity.

The two ends to be joined should be prepared carefully. The best method is by sealing off and blowing out, but in some cases this may not be possible. If the end of one tube is jagged, resulting from a break in the apparatus, and it is not practicable to close it and blow it out, the end should be smoothed by heating and working to a flat surface with a tool. This is done by removing protuberances with forceps or scissors, and then pushing the surface flat with the tool. The importance of working the ends of the tubes so that they are a good fit when pushed together cannot be overemphasized at this stage, since a good fit eliminates much work in the later stages, and overworking of glass leads to devitrification of the surface.

The working of the joint is carried out by first clamping the movable tube loosely in position so that the ends of the two tubes to be joined are nearly touching. The clamp should be left loose so that the two tubes can be pushed together later, but it should be tight enough to support the tube and not allow it to sag. The prepared ends of the tubes are heated, gently at first, with a hand torch, taking care to heat more or less evenly from all sides. Asbestos screens may be useful in joints of this type for protection of other apparatus. When the ends of the two tubes are hot enough to flow, the loosely clamped tube is pushed gently into place so that the glass is in contact all round the joint. The clamp is then tightened to prevent further movement. The glass is worked by heating and blowing in much the same way as described in the preceding section except that blowing is carried out by a blowing tube attached to one or other of the two tubes. In making fixed joints such as this it is undesirable to overheat the glass so that it becomes very liquid, since it flows under gravity. This flow should be kept to a minimum since it leads to uneven wall thickness.

The effects of this flow under gravity can be mitigated in a number of ways. If the tubing being worked is vertical it may be drawn down a little just below the join to reduce the wall thickness in this region; then, as the glass flows, this thin section will thicken so that a uniform wall thickness is achieved. Another method which can prove very useful to the amateur glass-blower is to make use of a spike to push the glass back to its original position. If the glass is heated until it is fairly soft, the point of the spike can be used to lift the glass which has dropped down. It is best to lift this glass above the position of the joint as it has to be worked to remove the local thickening of the wall

JOINING TUBES OF SIMILAR SIZE

produced by lifting, and during this working it will tend to flow down again. This also applies to a horizontal joint where the glass tends to collect at the bottom of the tube.

It is very often possible to produce a better finish to a joint if the glass is actually blown while it is in the flame, but if this is done great care must be exercised to blow very slowly and with short puffs rather than a long steady pressure. It is also virtually impossible to do this in a system which has a volume in excess of two litres since the pressure response is subject to too great a time lag and when the blowing is stopped the pressure inside the system may not drop for several seconds, which may lead to a hole being blown in the glass.

An alternative procedure is to hold the movable tube in one hand while the other hand operates the hand torch: the tubes are heated and pushed together as before, and the movable tube is supported and manipulated with one hand. With tubing over about 1 cm diameter, the movable tube needs little or no support most of the time, as the side not being worked on is usually rigid enough to hold it. Hence it is possible to use welding rod for filling up any holes in the joint. The blowing tube should be attached to the fixed tube, to reduce the weight on the movable tube to a minimum. It is convenient to have the tubes vertical with the movable one held below. This general method is usually preferable to the method involving clamping the movable tube, unless the latter is very heavy.

An elegant joint can be obtained by heating the joint strongly all round, and as it thickens moving the lower tube up; then the whole is blown out and the lower tube is moved down, so that a smooth joint is made. This procedure resembles the joining of two tubes by Method 1.

(3) *Joining Two Tubes both of which are Fixed*

This situation may arise in the repair of apparatus or in making a closed system. If the joint is a good fit no great difficulty is to be expected as the case is similar to that just described, except that the ends cannot be pushed together unless they are in fairly close contact, when they can usually be pushed together with a spike or some other tool when the glass is hot.

If the tubes are so placed that there is a gap between them it is possible to close this gap by various stratagems. One of the two tubes may be drawn down slightly so that their ends are brought into closer contact. If this tube is vertical it is best to make the point of drawing just below the proposed joint and to use forceps to hold the short end of the tube just above the point of drawing. This will provide the constriction to take up the glass which flows down under

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gravity. If the gap is not too great it may be filled in by building up a network of glass in it, using thin rod. This process is rather similar to darning the gap: a closely packed network of thin rod is built up in the gap and is then heated fairly strongly to fuse the individual rods together. The glass should not be heated too strongly while the original network is being built up, as strong heating very often leads to complete closure of the tubes. This method can be used for filling in irregularly shaped holes and finds many applications, particularly in the repair of apparatus which has been broken.

If the gap between the two pieces is too large for the above method to be attempted it may be possible to fit a sleeve of wider tubing over the two tubes and fuse its ends down on to them. This method inevitably leads to a very thick-walled joint, but the excess glass can be pulled off with forceps or by fusing a glass rod on to the thick part and pulling it out.

All joints, no matter how they are made, should be annealed with a relatively cool flame (p. 170).

When joints are made by the above methods, pinholes are very often left in them, particularly when they have been built up of thin glass rod. These pinholes are usually fairly easily visible when the glass is hot—at red heat they usually show up as fine orange lines running through it. They are sealed by placing a small spot of glass over their outer end using a thin glass rod for the purpose. The spot of glass is then fused in thoroughly. It is obviously more convenient to avoid such pinholes when the joint is made than to have to re-heat it later. It is desirable to wear glass-blowing spectacles when looking for pinholes.

Devitrified glass should be pulled out of a joint with forceps or a glass rod, and if this leaves a thin-walled joint extra glass can be built on using thin rod.

The aim in all these operations is to produce a joint with the same bore as the tube and with uniform wall thickness. Only practice will allow this to be achieved. However, with *Pyrex*, very bad-looking joints will hold together.

Joining Tubes of Different Sizes

There is very little basic difference in the procedures for joining tubes of different sizes and for joining tubes of the same size. The main difference lies in the preparation of the end of the larger tube: the end of this tube should be drawn down to a spindle and the spindle cut off in the flame. The closed end so produced should be heated locally so that when it is blown out the hole produced will be of the same size as the bore of the smaller tube. But it is usually better to

JOINING TUBES OF DIFFERENT SIZES

aim to produce a hole rather smaller than this, since it is a simple matter to open out the hole with a tool or to collapse the smaller tube down slightly; whereas it may be more difficult to close the hole in the drawn-down portion of the larger tube if this has relatively thin walls or uneven wall thickness. The above remarks apply equally to the use of a bench torch and to the use of a hand torch with fixed tubes.

When the joint is to be made using a bench torch it will usually be found easier to work so that the glass is blown through the smaller of the two tubes, when the joint is more easily visible. If the larger tube is used for blowing, the joint may be screened from view by it. When the novice starts trying to make this type of joint with a bench torch, difficulty may be experienced in rotating the two tubes of different

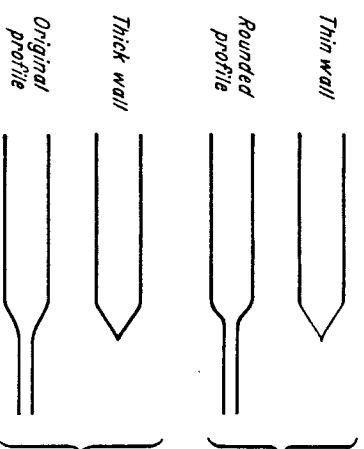


FIGURE 23. The effect of wall thickness on the joint, when tubes of different size are joined

bore at the same speed. The only remedy for this difficulty is practice, but it may be possible to avoid it by clamping the glass and using a hand torch. If only one tube is clamped, it should be the larger one.

The wall thickness of the taper produced by drawing down the larger tube can have an effect on the profile of the finished joint (FIGURE 23). If this taper has a wall thickness about the same as the unworked tube it will tend to retain its profile in working, but if it is thin-walled the joint will tend to have a rounded profile. This provides a method of controlling the type of joint produced. A rounded profile can be produced by starting with a uniformly thick rounded end.

T-Joints

The making of T-joints is probably the most common laboratory glass-working operation and therefore should be mastered thoroughly.

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There are two basic methods of making these joints, (1) by the use of a bench torch and holding both tubes, (2) by the use of a hand torch and clamping the tubes (or one of them).

(1) *The Use of a Bench Torch*

Once again the first step is the adequate preparation of the surfaces to be joined. First the side arm should be prepared, as for butt-sealing two tubes, by blowing out the end (p. 59). A hole is then blown in the side of the tube at the position required (see p. 59 for details). Care should be taken not to make this hole too large. Both ends of the tube should then be blocked since it is best to blow through the side arm so that the whole joint can be seen while blowing. Some workers, however, prefer to blow through the cross-piece.

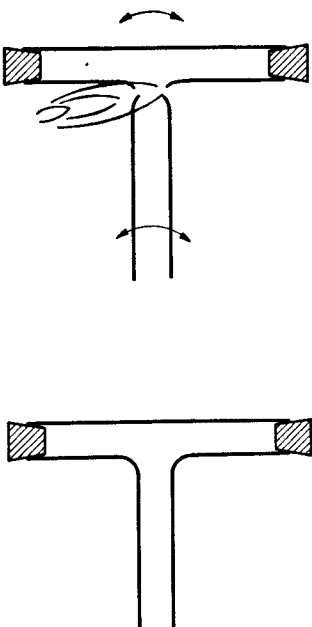


Figure 24. T-joints

The side arm is taken in the right hand and the cross-piece in the left and they are held in the flame so that the end of the side arm, and the area just around the hole in the cross-piece, are heated. This is done by directing the flame almost perpendicular to the side arm and swinging both pieces of glass through an angle of about 200° (FIGURE 24). These motions should be synchronized in order to obtain uniform heating. A great deal of practice is necessary before this can be done satisfactorily.

The two tubes are pressed lightly together when the glass is fused and the 'swinging' of the glass in the flame continued until the region of the joint is thoroughly fused. The joint is then blown through the side arm. This is repeated until the glass has been worked sufficiently for a good joint to be obtained and it should then be annealed.

The above procedure can be simplified in two ways. First, it is not necessary to swing the cross-piece in the flame during the preliminary heating before joining the tubes, although this operation accustoms

T-JOINTS

the two hands to synchronous working. The side arm only is rotated in the flame, which is meanwhile directed on the hole in the cross-piece. Secondly, a T-piece can often be made by working only a portion of the joint at a time.

The procedure for this type of joint is the same for all sizes of tubing. In some ways it is rather easier to carry out for relatively large sizes (about 2 cm) than for smaller sizes (less than 8 mm). In the latter case it is very easy to heat the cross-piece in such a way that it bends at the joint. This may be due to several possible causes: too large a flame; failure to hold the glass at the correct angle to the flame; or overheating of the glass in the vicinity of the joint, the heat being transferred by conduction.

A special technique for making T-joints, which does not distort the bore of the main tube if it is done correctly, is to make the joint by sucking: the end of the side arm, and a localized area on the main tube corresponding in size to the cross section of the side arm, are heated strongly until the glass is very soft. The end of the side arm is then placed on the heated area of the main tube and the heated area sucked up strongly into the side arm which must have a plug of cotton wool in it to catch any glass fragments. This produces a sound joint, but it requires very thorough annealing because of its very local heating.

(2) *The Use of a Hand Torch*

A casual study of any laboratory vacuum apparatus will show how commonly T-joints must be made with either one or both of the tubes **clamped**. Generally speaking this is a much easier technique than that previously described, as it removes the necessity for synchronized manipulation of the glass.

The glass is prepared in the same way as previously described, and a blowing tube is connected to some part of the system. The tubes are clamped in position and one of them left slightly loose so that they can be pushed into close contact at a later stage. The glass is heated in the vicinity of the joint by working around it with the hand torch and the two tubes are pushed together. The glass is then worked by collapsing it down and blowing out until it has fused together thoroughly. The joint is then blown out to the desired profile.

If difficulty is experienced with the glass flowing under gravity the same stragem may be used as described in the section on butt-joining two tubes (p. 62). If for any reason the tubes cannot be pushed together the two pieces should be as good a fit as possible before the joint is started and may be pushed into closer contact when hot with the aid of a spike, or if this fails the gap can be filled in with

BASIC GLASS-WORKING OPERATIONS

thin rod. This may leave pinholes in the joint and a careful inspection should be made while it is hot. When larger size tubing is used it may be found advantageous to blow the holes in the tubes rather smaller than their bore, since this provides a means of thickening the wall of the joint as it is blown out to a smooth profile.

When making a T-joint with a hand torch it is often best to clamp the cross-piece and fix a blowing tube to it, and to manipulate the side arm with one hand. As with the joining of two tubes (p. 63), the side arm will be supported by the rigidity of the glass on one side of the joint most of the time, unless it is of small tubing. This method is usually preferable to that involving clamping both tubes.

The construction of big T-pieces is described on p. 94.

Y-Joints

Joining two tubes in the form of a Y is most easily effected by bending one of the tubes to the desired shape and then making a normal T-joint at the bend. If the T-joint is made first and then one of the arms is bent, the resulting Y-piece will probably not be symmetrical and the finish on the surface of the glass may not be good owing to volatilization of some of its components during the making of the joint. This causes the surface to wrinkle on bending the tube.

The most convenient order of carrying out the various steps is to prepare the side arm first, bend the main tube and blow out the hole for the T-joint, and then make the joint before the glass has time to cool.

4 and 5 Way Junctions

Once the simple T-joint has been mastered the manufacture of 4 and 5 way junctions presents no difficulty: the operations described for making T-joints are just repeated the required number of times. The various side arms are all prepared as the first stage, and sufficient corks, or other means of blocking the tubes as they are joined on, must be at hand. The only other step which requires extra care is to see that completed joints are not overheated in making others, since if any of the side arms are unsupported they will bend under gravity. Adequate annealing is, as in all other joints, essential.

Working Capillary Tubes

Working capillary tubing requires a slightly different technique from the working of normal tubing. This is principally due to the small bore of the tube and the very thick walls. Generally speaking it is necessary to use a smaller, hotter flame for capillary than would be

WORKING CAPILLARY TUBES

used for normal tubing. Three types of operation will be described in this section, (1) joining capillary to wider bore tubing, (2) joining capillary to capillary, (3) capillary T-joints.

(1) Joining Capillary to Wider Bore Tubing

The only difference between this type of joint and a normal butt joint is in the preparation of the capillary tube: first the end of this is sealed off and a small fairly thin-walled bulb blown equal in diameter to the hole in the wide tube. The end of this bulb is then blown out and the joint made by the normal method. The flame must be directed so that the capillary is not collapsed. If the capillary has a very fine bore some difficulty will be found in blowing through it. This may be overcome by connecting it to a compressed air line while the small bulb is blown; the end of the bulb can then be pulled out with a rod and cut off to give the prepared end.

If for any reason it is found to be impossible to blow out the end of the tube, the cut end of a capillary can very often be opened out by using a mounted needle from a set of dissecting instruments as a tool; but this is not recommended if the tube can be blown.

(2) Joining Two Capillary Tubes

Owing to the fact that capillary tubing is used in many systems to cut down the gas space it is often desirable for the finished joint to have the same bore as the tubes. Therefore it is undesirable to prepare the ends by blowing out; it is much better to join the ends of tubes cut at a file mark. They should be heated until the glass is soft, and the tubes then held at an angle to one another. The two tubes are brought into contact by touching their edges together, and then brought into line ensuring that the bores of the tubes are coaxial. The joint is then finished by heating as small a length of the tube as possible, collapsing the bore down to about half its size and blowing it out to about twice its size. This operation should be repeated until a good joint is obtained. These joints require careful annealing because of the heavy wall thickness.

(3) Capillary T-Joints

The side arm for the T-piece should be prepared by blowing the end out of a piece of capillary as described above for joining capillary to wider bore tubing. The cross-piece is prepared by heating a short length of the capillary in the flame and blowing a slight symmetrical bulb. This should not be more than twice the original bore of the tube. A very small hot flame is used to heat one side of this small bulb, which is then blown out as for a normal T-joint. If difficulty is

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experienced here, it may be found advantageous to attach a glass rod to the point heated on the side of the small bulb and pull this out into a cone. The rod should next be cut away with the flame at the point of the cone, which can then be blown out much more easily. Lastly the joint is completed in the normal way and annealed thoroughly.

Fine capillary tubes may become blocked with water when blown into from the mouth; it may be desirable to put a drying tube in the blowing tube or to blow with a rubber bulb.

Blowing Bulbs in the Middle of a Tube

To ensure that the centre of the bulb lies on the axis of the tube from which it is made, the tubing, when in the flame, must be rotated continuously as described on p. 52; otherwise the glass will sag to one

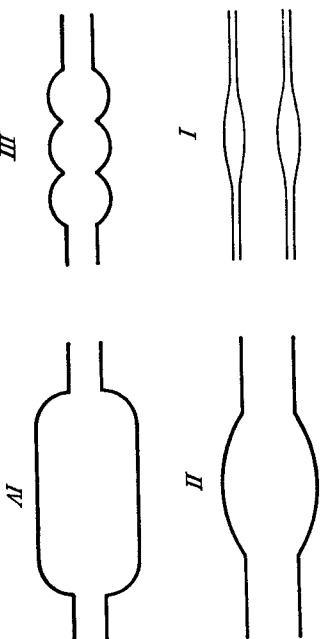


FIGURE 25. Blowing a bulb in the middle of a tube

side. Bulbs can be blown in the middle or at the end of a tube; the former operation is probably easier at first. A fairly large and relatively cool flame is used to heat a large portion of the tube, and as the tube is rotated in the flame the ends are pushed in to form a concentration of glass (FIGURE 25, *I*). The tube is taken from the flame and gently blown so that a slight bulge is formed (FIGURE 25, *II*).

The danger of distortion of the glass is much reduced when the blowing is done with a glassblower's swivel (p. 43): In this case the blowing tube is attached to the end of the tube—usually the left—which is held in an overhand grip. The process of accumulating glass is continued after the bulge has been re-heated, and the bulge is re-blown a number of times until enough glass has been accumulated for a final sphere to be blown. This may be blown by a series of puffs; the thinner parts blow out first, but they also cool first, and by blowing at the right moments a spherical bulb may be obtained. The

BLOWING BULBS IN THE MIDDLE OF A TUBE

tubing should be rotated when it is being blown into; this is facilitated if the lips are moistened before blowing.

Instead of accumulating the glass in one bulge, an alternative method is to make three bulges by pushing the glass together (FIGURE 25, *III*) and then reduce these to a single bulge by repeated blowing and shrinking. This reduces the time required to collect enough glass for the final bulb to be blown.

If the bulb to be blown is of radius much greater than that of the tube, a large tube should be joined into the small tube, as in FIGURE 25, *IV*. The three portions of tube should be coaxial. The large one is then shrunk and blown a number of times, and finally blown to the required bulb. It should be longer than the required radius of the bulb, as its length will decrease with heating and blowing.

To blow an elegant large bulb is very difficult. It is much simpler for the scientist to join a round-bottomed flask into the tube.

Blowing Bulbs at the End of a Tube

Blowing bulbs on the end of tubing may be found difficult because there is no portion of tubing to help to keep the bulb central whilst it is being blown. A round end is first made on the tube, which is

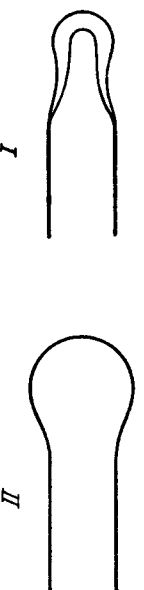


FIGURE 26. Blowing a bulb at the end of a tube

rotated continuously so that the glass thickens (FIGURE 26, *I*). The tube is taken from the flame, held near the vertical, and, with continual rotation, blown into until a slight bulb is formed (FIGURE 26, *II*). This is returned to the flame and collapsed and blown a number of times until enough glass has collected for the final bulb. When blowing this it is essential to rotate the tubing all the time, and to have a gradual air pressure, or a series of puffs, rather than a sudden off-pressure. The bulb must be watched as it is blown; if it is becoming off-centre it may be best to shrink it down and blow again. Beyond a certain size, however, a bulb cannot be shrunk, specially if it is thin.

Another method of blowing a bulb at the end of a tube is first to put a central spindle in it, and then accumulate glass just beyond the shoulder of the spindle by pushing the glass together with the spindle, and occasionally blowing through it. When enough glass has been

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collected, the spindle is drawn off in a very small flame, and the bulb is blown as described above.

To blow a big bulb a larger tube must be joined to the end of the tube and shrunk to give the necessary accumulation of glass. The beginner should be able to make useful small bulbs which will become quite elegant with practice. For research, however, it is best to fuse a round-bottomed flask to the tube.

Multiple Perforations in Bulbs

A bulb with a number of small holes in it is useful in bubbling a gas through a liquid. These holes are easily made: a small bulb is blown; a very small area of the bulb is heated strongly, and the glass is pulled out by touching it with the fine point of a glass rod, and then

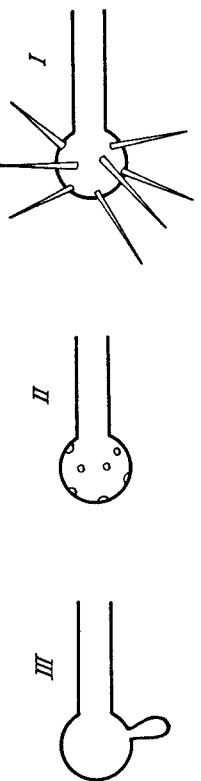


Figure 27. Making a number of holes in a bulb

drawing out the molten glass into a thin thread. This is repeated in a number of places, to give the result shown in FIGURE 27, *I*. The threads are then broken off near the bulb, and each hole so made is heated and rounded to give the finished perforated bulb shown in FIGURE 27, *II*.

An alternative method is to use a very small flame on the original bulb, and to blow a thin-walled bubble, as in FIGURE 27, *III*. A number of these bubbles are blown, then they are all broken off and each hole is smoothed off. If a bubble is broken accidentally, the hole can be blocked with damp asbestos paper so that more bubbles can be blown.

Flanging, Flaring or Bordering Tubing

A number of tools are useful in putting flanges of different kinds on the end of a glass tube. The operations are purely manipulative and require no blowing. Various tools which can be used in flanging are shown in FIGURE 4; perhaps the triangular-shaped tool is one of the most useful. Carbon rods are also useful (p. 33), and they have the advantage that they do not mark the glass; an overheated metal tool

FLANGING, FLARING OR BORDERING TUBING

sometimes leaves a small deposit of metal or oxide on the glass. The general method used in flaring is to heat the end of the glass tube over a length of 2-3 times its diameter, and when the glass has become soft and thickened (FIGURE 28, *I*) the tool is inserted and the tube rotated. The glass should be rotated fairly quickly backwards and forwards by rolling it between the forefinger and thumb of the left hand; or normal rotation may be used (p. 52). The glass should be *rested* on the tool, and the action should be gentle. The spinning, as this action is called, should be done with the flame impinging on the base of the cone formed (FIGURE 28, *II*). For a standard taper, a hexagonal

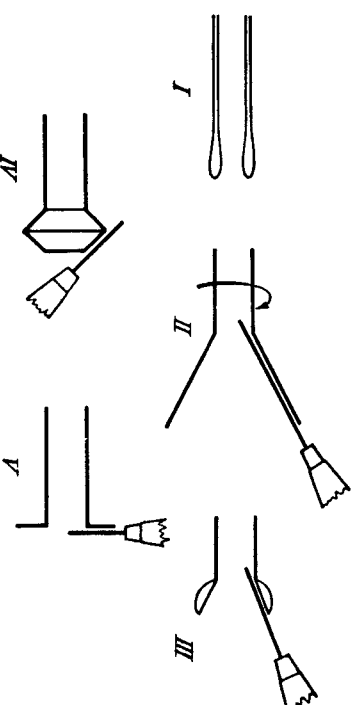


Figure 28. Some methods used for flanging, flaring or bordering tubing

reamer is used (p. 31). Lubrication is desirable to prevent the glass becoming twisted round the reamer. Various ways of using tools are shown in FIGURE 28, *III*, *IV* and *V*.

Internal Seals

Internal seals can be made by a variety of methods, which will be described; some indication will be given of situations in which they might be useful. Internal seals are also called ring seals.

Method 1

In this method a single piece of tubing is sealed into the larger tube. The first step is to form a slight ridge around the inner tube, at the point where it is to be sealed, by rotating it in a small hot flame and pushing its ends inwards slightly. If this is done correctly a symmetrical ridge should be formed.

The outer tube is then closed to a round end and a hole blown

BASIC GLASS-WORKING OPERATIONS

centrally in this, large enough for the internal tube to be inserted but small enough to prevent the ridge on this tube passing through. The outer tube should be longer than the inner so that the open end may be closed by a cork or rubber stopper (FIGURE 29, I); the blowing is carried out by blowing through the projecting inner tube. It may be more convenient to close the other end of the outer tube with a spindle. This gives a useful handle.

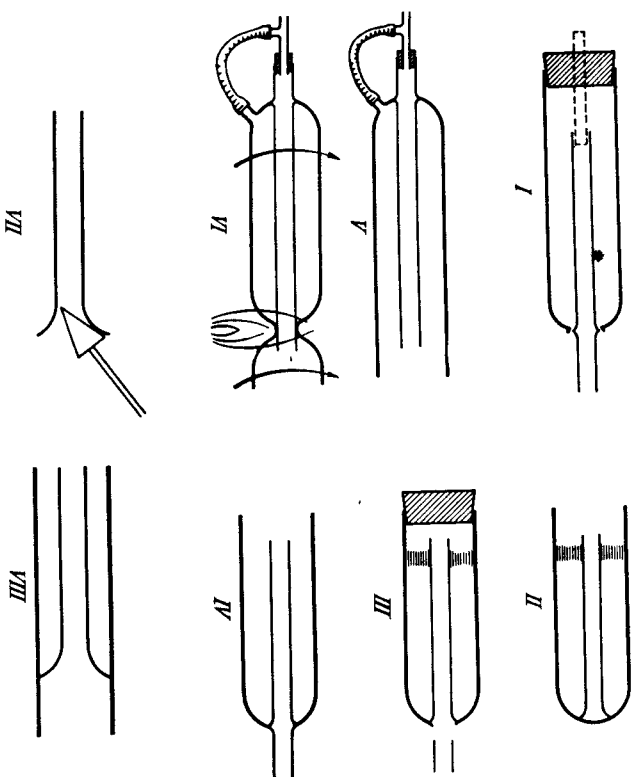


FIGURE 29. Various methods for making an internal seal

The glass is then heated by rotation in the flame so that the outer tube is fused to the ridge formed on the inner, and finally the joint is blown until a good-shaped joint is obtained. To get a good-shaped shoulder it is necessary to direct the flame mostly against the smaller tubing. The heating and blowing should be repeated until good fusion of the two tubes is achieved. Any pinholes present can be sealed as before by the application of a small spot of glass from a piece of rod. These spots of glass should be fused in thoroughly.

The unsupported internal tube will sag under gravity during the above operations, but if the rotation of the work is sufficiently uniform it can be centralized as the glass cools by allowing it to drop

INTERNAL SEALS

to its central position and keeping it there by rotation. Alternatively it can be supported on a rod held by the cork blocking the end of the outer tube. If this latter method is used care should be taken to see that the flow of air from the inner to the outer tube is not hindered since it is essential for the air pressure to be equal in the two tubes at all times.

Internal seals require more thorough annealing than the more simple joints, and if any pinholes require subsequent repair great care should be taken in reheating the joint.

Method 2

In this method the internal tube is in two pieces and there is less likelihood of making a joint with pinholes in it. The first step is the preparation of the internal tube: a length of tubing is drawn off, the end blown out and polished smooth in the flame. The projecting part of the internal tube should be prepared in the same way. The outer tube should be prepared by closing one end to a round end.

The joint can be made most easily if the internal tube is supported almost in its final place by pads of asbestos paper or by a glass holder, but whatever support is used it must allow the air pressure to be the same on both sides of the internal tube. The inner tube is put in place and the rounded end heated. When this is starting to collapse inwards the internal tube is pushed forward slightly so that its prepared end comes into contact with the heated end of the outer tube. This glass is worked thoroughly by heating and blowing until a good fusion between the two parts is achieved (FIGURE 29, II). The end of the tube is allowed to cool slightly and then, with a fairly small flame, the area of glass covering the internal tube is heated and blown out. The other end of the outer tube is then blocked with a stopper (FIGURE 29, III) and the remaining part of the inner tube joined on as in a simple butt joint (FIGURE 29, IV). Once again it is important that the air pressure should be the same in both inner and outer tube and that the tube should be rotated uniformly. Finally the joint should be annealed.

These joints can be made with the internal tube unsupported by inserting the inner tube only after the outer tube is hot enough for it to stick firmly and then proceeding as above. This is rather more difficult and is seldom as satisfactory as wedging the inner tube, although at times one is tempted to avoid this added step.

This method is not on the whole satisfactory: with soda glass, the cold inner tube will probably shatter, and if a borosilicate glass is used to avoid this, the difficulty of getting the inner tube central still remains. A better method is the following: the outer tube is

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rounded off and pulled to a slight taper. The inner tube is flared slightly so that it fits into the taper at the desired point and is thereby centralized. The tubes are fitted together when cold, and the operation is continued as before.

Method 3

The third method of making internal seals is not so widely applicable as the previous two, but it is very useful for certain joints. It is best described in terms of one of its uses—in the manufacture of a Liebig condenser.

Let us assume that the condenser is in the state shown in FIGURE 29, *V*: the first internal seal has been made by one of the two methods described above, and a side arm has been put into the water jacket. A connection should be made with rubber tubing, so that the water jacket of the condenser is connected to the inner tube, and then the outer tube is heated with a hot narrow flame to collapse it down on to the inner tube for the internal seal. When the tube has collapsed on to the second tube the heating is continued and both tubes drawn off together (FIGURE 29, *V7*). The system is then blown slightly to give a good round profile to the end of the outer tube; when this has been done the end of the inner tube is blown out and the joint finished as in Method 2. Another method of performing this operation is described on p. 130.

Method 4

This method involves the fusing of a flared-out narrow tube into a wider one. This type of seal has its uses in pieces of apparatus such as McLeod gauges. The first step is to flare out the end of the inner tube, using a flaring tool, until it will only just pass inside the wider tube. This is done by rotating the tube in a flame and using the flaring tool as shown in FIGURE 29, *VII*, or as described on p. 73. If a very large flare is required it will probably be necessary to fuse a wider tube to the narrow one as the initial stage in flaring. When the flare has been made to fit it is packed in place with asbestos paper or held with a rod, and the outer tube is collapsed down on to it (FIGURE 29, *VIII*). The outer tube is blown out again and the process repeated until a good fusion of the two pieces is achieved. This type of seal requires particularly careful annealing.

Use of the Hand Torch

Generally speaking it is best to make internal seals using a bench torch and rotating the glass in the flame. If this is not practicable it is possible to work with a hand torch with the glass clamped if Method

INTERNAL SEALS

I is used: in this case a rim of glass, left by the blowing of the hole in the outer tube through which the other tube passes, can be heated with a hand torch and pushed in against the walls of the inner tube with a spike. This can be done also in cases where the formation of a ridge of glass on the inner tube has been impracticable. Once an airtight joint has been made by pushing the glass into place, the hand torch can be used to fuse the two tubes together by working round the joint. As before it is essential to have the blowing tube connected to both inner and outer tubes.

One great difficulty with this type of work is the force of gravity acting on the inner tube: If this is unsupported it is very difficult to obtain a good-looking joint, and even if it is supported the results are not always satisfactory, owing to non-uniformity of heating. It is best to clamp the tubes vertically. The inner tube may be held in one hand. Difficulties may possibly arise if the outer tube has thin walls, because any holes will enlarge as the glass is worked. In this case they may be filled with rod as in the other cases described. Care should be taken in inspecting these joints for pinholes and annealing the work when completed.

Thin Glass Windows

In recent years the need has arisen for very thin glass windows capable of withstanding a vacuum yet thin enough to transmit α and β particles. S. ROSENBLUM and R. WALLEN (1945) have described the following method for putting very thin windows into capillary tubing of up to 2 mm bore. The process is very simple but may require some practice before the precise conditions necessary for a satisfactory window are achieved.

The end of the capillary tube is heated until the glass is soft, then before it has time to cool it is touched on to the surface of a thin bubble of glass and a slight suction applied. This forms the window into a concave shape and draws it slightly down into the capillary, whose ends then protect it from damage. The bubble of thin glass should be thin enough to show interference colours. These windows will stand a vacuum provided atmospheric pressure is on the concave side of the window. If they are subjected to a pressure difference in the other direction, failure occurs due to the reversal of curvature.

A method for making larger thin windows has been described by J. T. LLOYD (1949). For windows in the end of a tube he advocates the blowing of a thin-walled bulb at that end and then drawing it into the tube to form the window by the following method: the thin-walled bulb is blown at the end, the wall of the tube is then heated all round at a short distance from the end until the glass is fused, and then the

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bulb is immediately re-heated and sucked inwards. The bubble is drawn into the tube and is fused to the walls in the region where the tube was heated. This gives a relatively robust window capable of withstanding a vacuum and protected by the projecting ends of the tube.

Lloyd also gives details of a similar method for the formation of windows in the sides of tubes. The bulb is blown on the side of the tube in the same way as it would be in the first stage of making a T-joint; it is then heated and sucked in. Lloyd's method requires more practice than the other method before satisfactory results can be obtained, but it has the merit of being more versatile.

Spinning Out Feet

The stands on measuring cylinders, wine glasses, etc., are called 'feet' by the glass-blower. They are made by an operation similar to flanging: the glass must be thick enough for spinning out. A piece of 3-cm diameter tubing about 10 cm long, with spindles (spears) at

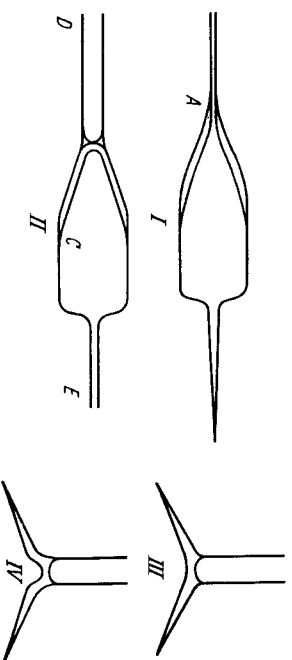


FIGURE 30. The production of 'feet' for vessels

each end, is prepared. One end is then heated in a large, relatively cool (soft) flame until it thickens, and, by pushing with the spindle and by blowing, the shape shown in FIGURE 30, *I* is produced. The spindle at *A* is removed and the end closed. This closed end is then joined to the closed and rounded end of the tube for which the feet are required. The two ends are simply brought together (FIGURE 30, *II*), and, by blowing through both *D* and *E* and slightly pulling, a good joint is made. The tubing is next cut at *C*; and the feet are worked to shape with a rounded tool (FIGURE 4, *G*), using the same operations employed in flanging (p. 72). It is important, however, to get the glass very hot and soft, so that the feet can be spun out in one operation, and not to push too hard with the tool, to avoid producing buckled

SPINNING OUT FEET

feet. As the hot glass is rotated horizontally, the tool should be slowly brought from the horizontal to the vertical. The spinning out should be done along the whole width of the glass to produce the shape in FIGURE 30, *III*; if only the outside is spun out the shape in FIGURE 30, *IV* is produced.

Tapering Glass Tubes

Glass tubing may be tapered by two methods, depending upon the wall thickness required. For a thin-walled taper, a central spindle is first put in the tube, and the shoulder is heated in a large soft flame. The tubing is rotated and slightly collapsed, and then taken from the flame and pulled out to the required taper, with constant rotation. This will give a fairly thin-walled taper, conical in shape; but by slightly pushing in the ends while heating the shoulder a thicker wall can be obtained.

For a thick-walled taper the operations are the same except that the tubing at the shoulder is shrunk down and blown out a number of times to accumulate glass. When enough glass is present, the shoulder is made hot, slightly blown out, and then drawn down out of the flame.

To taper a tube from one size to another, two tubes of the desired sizes are joined and worked to a taper at the joint.

When making tapers to fixed dimensions it is convenient to draw the required taper on asbestos paper and work the glass to match the drawing.

Sealing-in Sintered Glass Discs

The glass disc to be sealed into a tube must be made of glass having the same thermal expansion coefficient as the tube. Discs of different porosities can be obtained; they consist of a large number of particles of glass sintered together so that numerous channels remain. When



FIGURE 31. Sealing a sintered disc into a tube

working with a disc it should be warmed up carefully. One end of the tube into which the disc is to be sealed is closed with a rubber stopper carrying a tube *A* (FIGURE 31). The disc is rested against this tube, and should just fit into the outer tube. If that is too big, a constriction must be put in.

BASIC GLASS-WORKING OPERATIONS

The tube and disc are carefully warmed until the flame is yellow; one side of the tube is then heated strongly and attached to the disc. The disc is next made perpendicular to the tube by pushing with a rod through the end *B*, and manipulation with *A* if necessary. The tube is allowed to cool slightly so that the disc is firm, and this is then attached again to the tube by strong local heating opposite the first point of joining. The whole tube is then shrunk down on to the disc. It will probably be necessary to blow the tubing out a few times, blowing through both ends. The tubing can finally be pulled down to a uniform diameter. The joint should be annealed with extreme care, preferably in an oven.

Mending Cracks

Unfortunately, mending cracks in laboratory glassware is one of the more common tasks. Cracks usually occur in glassware for one of two reasons—either failure to remove intrinsic strain from the glass by annealing, or else straining of the glass by careless handling of the apparatus itself. Whatever the cause of the crack the procedure for repairing the damage is the same.

The first stage in mending cracks and the one requiring most care is the initial heating of the glass: If this is done carelessly the crack will almost certainly spread much further. The glass should be warmed up slowly with a small flame, very little oxygen or air being mixed with the gas. This small flame should be played over the surface of the glass in such a way that it does not remain directed at any one part. The whole region surrounding the crack should be heated and not just the immediate area, particular care being taken if the crack is near an internal seal to see that this is heated evenly. As the glass warms up the size of the flame and the proportion of air or oxygen in it can safely be increased until the glass becomes soft and there is no danger of further cracking. For very hazardous repairs it is desirable to heat first with hot air, and then a flame with no air or oxygen until the article is covered with soot, which probably assists the uniform distribution of heat. When the soot starts to burn off, the most dangerous part of the operation has been reached, in our experience.

When the glass is soft the flame should not be made any hotter—a relatively cool flame is much more effective for sealing cracks. The flame is then played on to the crack, starting at one end, and the glass heated gently. It should be possible to see the crack sealing up under these conditions and the flame should be moved along it as it seals. If too hot a flame is used the crack will open as it is heated because the glass at its edges will become sufficiently hot to draw back under surface tension before the inner parts of the crack have had a chance

MENDING CRACKS

to fuse together: this is then a progressive process and the crack can only be sealed by filling in with glass rod or pushing the sides together with a tool.

When the crack has sealed, a hotter flame can be used and the glass worked by heating and blowing until the crack has been completely sealed. It is necessary to keep the glass in the region of the original crack, and all the surrounding parts, at a dull red heat for a short while (for *Pyrex*) and then to cool it slowly to allow sufficient annealing time. Care must be taken with this general annealing of a fairly large area of glass if further cracks are to be prevented.

Extremely unpleasant cracks can be sealed in *Pyrex* glass. With soda glass, however, the sealing of cracks is a hazardous undertaking, and the whole apparatus may be lost. Cracks can be sealed in *Monax*.

Detection and Removal of Leaks and Holes

Holes in awkward places are often difficult to find when one is assembling complex apparatus. If the joint in question is put in a luminous flame, or a flame put round the joint, and the apparatus blown into, the air will rush out of the holes and blow the flame away in a blue jet. The hole, when found, can then be repaired with thin glass rod (p. 58); or, if it is very big, it can often be filled in with a piece of glass. When a piece of glass falls out of an apparatus, it can be fused to a rod, to provide a handle, and put back again with the help of thin glass rod.

These methods work well with *Pyrex* glass, but not with soda glass, which cracks too easily on re-heating. A hole which has been mended with blobs of glass from a rod needs to be fused down and blown out a few times, to make it more homogeneous. Fusing down and blowing out glass often closes a small hole—this method works better with soda than *Pyrex* glass.

Small holes and leaks can often be found by covering the outside of the apparatus with soap solution and noting where bubbles appear when it is blown into. Very small holes can be found with a high frequency discharge: the apparatus is evacuated with a rotary pump (or a good water-pump) and the electrode of a Tesla coil is brought near it. A discharge appears inside the apparatus, and if the electrode comes near a hole or fine capillary in the glass, the discharge passes through it and a very bright spot of light appears at the leak. With a water-pump vacuum the method is not so sensitive, because only a poor discharge is obtained in the apparatus. When the pressure in the apparatus is about 0.1 mm, very fine leaks can be found.

However, the Tesla discharge may blast a hole in the tubing, and is not useful for glass-to-metal seals, because it goes straight to the

BASIC GLASS-WORKING OPERATIONS

metal. If the seal is covered with cotton wool soaked in carbon tetrachloride or alcohol, this will pass through any leak, and should produce a greenish or bluish colour in the Tesla discharge. This method, however, is not reliable.

The minute capillary leaks found with a Tesla coil should be plugged with a blob of hot glass; the whole joint then needs heating and annealing. When assembling complex apparatus, parts which will become inaccessible to the flame must be tested for leaks before they are put in their final positions.

Grinding Glass

Glass surfaces which must fit closely are usually ground to shape. The most common example is the cone-and-socket joint: the surfaces are first worked to an approximate fit with fairly coarse carborundum (about 100 mesh) which is put between them and wetted. The cone is rotated by hand or in a lathe or drill chuck, or by a special machine, until the surfaces meet closely—the carborundum must be kept moist all the time. The coarse abrasive is then washed off, and the grinding continued a number of times with progressively finer abrasives. Various possible abrasives are mentioned on p. 39.

Surfaces which require regrounding, such as those of stopcocks, usually fit fairly well, and a fairly fine grade of carborundum can be used for the grinding. Stopcocks must not be ground too much, or the holes in the plug will no longer meet those in the barrel.

A flat surface can be put on the end of a tube by grinding. Such a surface may be required when a flat plate has to be cemented on to a tube. The flange is ground against a flat metal or glass plate using wet carborundum as an abrasive. The flange must be kept steady and not rocked, and is ground with a rotating motion of the whole tube about a point outside it. Progressively finer grades of carborundum are used. An operation of this kind is carried out very easily when a lapping wheel is available (p. 44).

Releasing Frozen Glass Surfaces

Ground surfaces are often found to be jammed together. If the parts are robust, a sharp tap will often release them. A glass stopper stuck in a bottle is often released by tapping with another glass stopper, or something solid. When the substance in the bottle has cemented the stopper in position, the joint may be put in a suitable solvent. Frozen surfaces are often more easily released after the joint has been immersed in a detergent solution for some time.

Fragile joints which cannot be struck may often be released by

RELEASING FROZEN GLASS SURFACES

quickly warming the outer member. This may be done with hot water, or, less safely, with a flame. If the surfaces are made too hot they are melted together and the apparatus must be discarded.

Polishing Glass

For many years, the standard method for polishing glass in the laboratory has been by rubbing the glass over a sheet of brown paper covered with a mixture of flour emery and a solution of camphor in turpentine. C. V. Boys (1927) has stated that he can find no reason other than tradition for the use of the solution of camphor in turpentine rather than pure turpentine for lubricating grinding media.

When the best possible polish has been achieved with the above mixture, rouge and water should be used for improving it. During the polishing only a fairly light pressure should be applied and the specimen should be moved fairly rapidly. If the polishing is being done manually, pressure should be applied evenly over the whole surface being polished. It is possible, however, to mount small samples in pitch and use a rotating polishing surface.

Cerium oxide (*Certrouge*) has been recommended as being superior to rouge as a polishing material (*J. sci. Instrum.* 22, 1945, 224) and special cloths are marketed impregnated with cerium oxide. These are useful for polishing glass and other materials such as fluorite. For details of the methods of polishing optical surfaces, works on that subject should be consulted.

Drilling Holes in Glass

The problem of drilling holes in glass has occupied the attention of scientists since the days of Faraday and many methods have been described. Articles reviewing the subject have been published by C. V. Boys (1927) and more recently by P. GRODZINSKI (1953).

The method which has been used most widely is probably that involving a copper or brass tube in conjunction with a grinding mixture of carborundum and water; a slotted copper rod is also effective for small holes. To obtain the best results by this method a fairly high drill speed should be used and the copper tube drill should be raised from the cut periodically to allow a fresh supply of the grinding medium to enter the hole. It is also desirable to have some slots in the end of the tube to hold the grinding medium. The pressure on the drill should be fairly light, as too great a thrust may splinter the glass and may also slow up the cutting by preventing the grinding medium from getting between the rotating drill and the glass. Grodzinski favours the use of diamond powder in place of the carborundum,

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but in most laboratories this is not obtainable. It is easy with a tube to obtain holes 3 cm in diameter.

Many variations of the shape of steel drills for glass have been described—from Faraday who used the ground end of a triangular file to C. V. Boys who designed a special drill of tool steel—while other workers (B. BROWN, 1927; N. G. HEATLEY, 1938) advocate the use of a normal twist drill. These drills are used in conjunction with some lubricant—usually the traditional camphor in turpentine, although pure turpentine or other mobile liquid seems to be as good, while Heatley states that dilute sulphuric acid is effective. To avoid splintering, the hole should be drilled from both sides.

Grodzinski, commenting on the use of sintered carbide drills, states that high speed is necessary for their use and strongly favours diamond drills. He gives a full account of their use and gives details of the mounting of diamonds for this purpose. Probably the simplest method of preparing a diamond drill for small diameter holes was described by Heatley (1938) who broke off the end of the elongated eye of a sewing needle and soldered a small diamond chip into this. He was able to drill quite deep holes, of diameter less than 1 mm, in glass with this type of drill.

Cementing Glass

Standard cones and sockets of metal and glass can be joined with *Piclen* or *W* wax; such joints are vacuum-tight. Windows can be put on metal apparatus with these waxes. *Piclen* is a black wax of low vapour pressure, and flows freely at about 80°C. Three useful *Apiezon* waxes are *W*, *W100* and *W40*, which are applied at temperatures of 100°C, 80°C and 40–50°C respectively. When a large glass socket is waxed to a metal cone, the strain set up on cooling can easily crack *W* wax, but *W40* is satisfactory for large joints. These waxes can also be used to seal two glass surfaces together, and a vacuum-tight joint can be obtained without ground surfaces.

Aero Research Ltd make two resin cements, *Araldite type 1* and *Araldite 103* and *Hardener 951*. They are epoxy resin cements, the former being thermal setting and the latter cold setting, and make transparent seals. To use *Araldite type 1*, the surfaces are cleaned, degreased and heated to 100°C, and the cement in solid form is applied. It melts and runs smoothly over the surface. The surfaces are pushed together and held together by a clamp while the whole is heated for a curing time dependent upon temperature, varying from 10 minutes at 240°C to 7 hours at 140°C. These are minimum times, and should be increased for stronger joints.

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To use *Araldite 103*, 8–10 parts of *Hardener 951* are added to 100 parts of *Araldite*—they are both liquids. The mixture is stirred and allowed to stand before use for 20–30 minutes at room temperature. The pot life of the mixture is about 3 hours at 20°C, so it is made up as required. The glass is degreased and the cement applied with a spatula or brush as a thin, even coating on both surfaces. The surfaces are immediately joined and clamped lightly. The cement cures at a rate depending on temperatures. 16–24 hours are required at 20°C and 10–20 minutes at 100°C. It is claimed that *Araldite* can be used for vacuum-tight, glass-to-glass seals.

Mica windows can be fused to glass using a special powder of a lead borosilicate glass softening at 450°C (J. S. DONALD, 1942). A suitable glass for solder seals of this kind is *BTH No. 93*.

Silvering of Glass

Glass surfaces may be silvered by chemical precipitation of silver on to them. This is most conveniently done by reduction of a silver salt; the speed of precipitation depends on the reducing agent used. The glass must first be cleaned: it is degreased with benzene, washed with acetone and water, and left in 'chromic acid' (see p. 49) for an hour or more at room temperature. It is then washed with tap water, followed by distilled water. Some workers prefer to then clean again in concentrated nitric acid, followed by the water washings. The glass must on no account be allowed to dry after these washings.

The speed of precipitation of silver required depends on the surface to be silvered. If the rate of precipitation is too great, the silvering agent may be spent before it can be removed from the surface, and a 'bloom' will appear because the silver is in contact with the spent solution. This is a discolouration of the silver. Thus if a rapid silvering agent is used for Dewar flasks, it may be difficult to avoid spoiling the silvering.

The silvering agent usually consists of two solutions, sometimes three, which are mixed immediately before use and poured on to the object to be silvered. Various recipes have been proposed (I. C. GARDNER and F. A. CASE, 1931). The following are useful for slow silvering (R. B. SCOTT, J. W. COOK and F. G. BRICKWEDDE, 1931):

Solution A—50 gm silver nitrate in 2l. distilled water

Solution B—90 gm potassium hydroxide in 2l. distilled water

Solution C—(80 gm cane sugar in 800 ml. distilled water) plus (100 ml. ethanol and 3.5 ml. nitric acid sp. gr. 1.42).

Solution A should be kept in the dark, and C improves with age. The solutions are used in the proportion A : B : C = 16 : 8 : 1 by volume. To use, concentrated ammonia is added to A drop by drop,

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with stirring, until the precipitate first formed is redissolved. The requisite amount of B is next added, and a dark brown or black precipitate forms. (But if enough ammonia has not originally been added to A, a yellowish-green precipitate will be formed; the solution should be discarded in this case). Ammonia is then added until the brown-black precipitate is almost redissolved; and to the resulting solution the correct amount of C is added to give the final, prepared solution. The intermediate solution prepared from A plus B can be used immediately, or kept up to an hour, after which it loses its activity. If allowed to stand for about half an hour it will become practically clear.

To silver a mirror a bath should be used just larger than the article to be silvered. The glass should be cleaned and under no circumstances allowed to become dry. The prepared solution is poured over the glass and the bath is rocked steadily. The silvering must be stopped at the right time or the mirror will be tarnished. The end-point is when a flocculent precipitate is formed. The time for this to happen depends on a number of factors: old silver nitrate solution deposits silver more slowly; temperature alters the rate, and a test run to find the approximate time may be desirable. To see how the silvering is proceeding one may take the mirror out of the solution, but it should never be exposed to the air for more than 1 or 2 seconds. The rocking during the silvering prevents precipitate adhering to the surface and spoiling the mirror.

The mirror can be silvered a second time if a thicker coating is required. Brighter and thicker coats are obtained by using a more dilute solution twice instead of a more concentrated solution once.

Front-silvered mirrors can be polished with optical rouge on a pad of chamois leather over cotton wool. Care is necessary because it is easy to polish completely through the silver. Back-silvered mirrors can be protected by first varnishing the silver and then painting the varnish. Front-silvered mirrors can be protected with a thin coat of lacquer, such as a dilute solution of cellulose nitrate in amyl acetate. The lacquer should not be too thin, or interference colours may be produced.

In order to silver Dewar flasks the jacket is cleaned and filled about two thirds full with the prepared solution. The Dewar should be rolled or shaken during the silvering. The end-point is determined by occasionally pouring out a little of the solution, and when a flocculent precipitate is formed the solution should be removed from the flask as quickly as possible to prevent blooming. The flask is then cleaned and dried. If the glass is first etched with a dilute solution (about 0.7 molar) of hydrofluoric acid before its final cleaning, the

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stability of the silvering with respect to peel-off is increased, and spots are less likely.

A simple method for filling and emptying Dewar vessels is mentioned on p. 123. F. E. J. OCKENDEN (1938) has given some hints on silvering glass, and E. A. BAKER (1931) has described a method for electrolytically removing silver from one face of a small mirror. P. B. G. UPRON and E. F. G. HERINGTON (1950) have described a method for depositing a heavy silver film by reduction of a silver ammonionitrate solution by hydrazine sulphate. The deposition of silver other than on the desired surface is prevented by pyridine. The effect of the pyridine is due to the presence of surface-active impurity, and the process becomes controllable by adding cetyl pyridinium bromide to 'inactive' pyridine.

Depositing Copper on Glass

Copper can be electro-deposited on to a silvered article: the plating bath consists of 10 gm of copper sulphate in 100 gm water. The article is taken from the silvering bath, washed, and put wet in the plating bath—on to a copper cathode so that electrical contact is made to the silver. A copper anode is placed 1–2 cm above the article. The electrodes should be about the same size as the article. The current density should be about 0.05 amp/cm², and in 2–4 minutes a thin copper film should be made. If the current density is too high a granular deposit will be formed which will rub off, and if it is too low a non-uniform deposit will be formed.

Copper can be chemically deposited in the same way as silver, using phenylhydrazine as the reducing agent. F. D. CHATTAWAY (1908) found this to give good results with blown glass but not with glass that had been polished. For plane surfaces E. A. H. FRENCH (1924) obtained good results with hydrazine sulphate.

Many metals can be deposited on glass in a vacuum, by vaporizing the metal and condensing it on the glass. J. YARWOOD (1955) has given a general account of this method.

Metallizing Preparations for the firing on of Metal Layers on Glass

Preparations of silver, platinum, gold and palladium are available which consist of a suspension or solution of the metal, or a compound, in an organic material; adhesion-promoting materials are also present. The cleaned glass is coated with the preparation by brushing, spraying or dipping. It is then heated fairly slowly to a high temperature (the firing temperature), when the organic materials are oxidized or volatilized and the metal becomes strongly bound

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to the surface. The preparations mentioned here are all made by Johnson, Matthey & Co.; they also supply thinners for them.

A glass surface to be silvered by a firing-on process should be cleaned with soap and water or detergents. For metallizing with platinum, gold and palladium, improved results are sometimes obtained with glass which has been stored when, after cleaning, it is immersed in 5 per cent acetic acid (cold) for six hours. The special preparation is then applied. If necessary it is mixed before use with a suitable thinner. The preparation is allowed to dry; this may be accelerated by warm air. Dust should not be allowed to settle on the surface before firing.

After drying, the coated glass is put in a furnace below 100°C and then slowly raised to the firing temperature. The exact firing temperature is best found by experiment, but data provided by Johnson, Matthey & Co. give good guidance. A temperature 20°C below the softening point (Mg point) of the glass is often used. The furnace is kept at the firing temperature for about 20 minutes, and well ventilated so that the organic constituents burn away. With palladium, however, oxidation of the metal occurs at 400°C and the supply of air to the furnace should be restricted during the final stages of the firing operation. The furnace can be cooled at any rate which does not introduce permanent strain into the glass (p. 19). The cooling to the lower annealing temperature of the glass should therefore be at the same rate as that for annealing the glass, or at a lower rate.

Some further details concerning the particular metals are given below.

Silver

The silver pastes give a thickness of silver of about 25–30 microns after a single application and firing. Successive applications and firings can be made up to a certain limit. *Silver Paste Y.40* gives a thick and strongly adherent silver film. One application of *Silver Paste X.351* followed by one of *X.353* gives a thick, matt, silver deposit. Several other silver pastes are made by Johnson, Matthey & Co. *Silver Paste O.182* is for mounting quartz crystals. Their liquid burnish silvers are much less viscous than the pastes, and with one application give a silver film about 3 microns thick.

The fired-on silver films on glass can be soft-soldered. They should be 'tinned' with a solder containing some silver; ordinary soft solder can bring about complete dissolution of the silver film. *Matthey Ceramic Solder No. 1* is a suitable solder for the films. A resin-type flux should be used. Tinning can be carried out by dipping into the molten solder.

THE FIRING ON OF METAL LAYERS ON GLASS

Platinum, Gold and Palladium

The films of these metals are much thinner than those of silver. A single application of *Liquid Bright Gold* (12 per cent), followed by firing, gives a gold layer of about 0.12 microns thick; one application of *Liquid Bright Platinum* gives a layer of about 0.08 microns; one application of *Liquid Matt Platinum* a layer of about 0.35 microns; and one application of *Platinum Paste* gives a layer of about 2 microns. These are too thin for soldering, except with considerable difficulty. Thicker films can be built up by successive applications and firings. The *Liquid Bright Platinum* preparations give brilliant mirror-like surfaces on glass; the films deposited by the *Liquid Bright Golds* are useful as optical reflectors and filters. Films of gold on glass can be prepared that reflect nearly all the infra-red radiation and transmit most of the visible spectrum.

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